



FIGURE 26.22 The Hubble Ultra Deep Field (HUDF), obtained by combining images from ACS and NICMOS onboard the HST. This HUDF image contains an estimated 10,000 galaxies covering a region 3 arcmin square (about 1/10 the size of the full moon) in the constellation Fornax. Some of the galaxies in the HUDF are so far away that we are seeing them less than 1 Gyr after the Big Bang. [Courtesy of NASA, ESA, S. Beckwith (STScI) and the HUDF Team.]

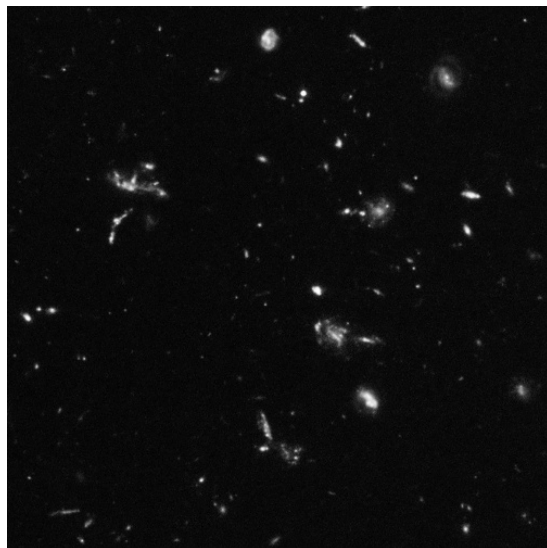


FIGURE 26.23 A close-up of a portion of the HUDF image shown in Fig. 26.22. Many of the galaxies in this image are extremely distant—and therefore extremely young. [Courtesy of NASA, ESA, S. Beckwith (STScI) and the HUDF Team.]

of work remains to be done before we can hope to fully understand the complex histories of galaxies, but it appears that many of the basic ingredients of an evolution theory may be in place.

SUGGESTED READING

General

- Barnes, Joshua, Hernquist, Lars, and Schweizer, Francois, “Colliding Galaxies,” *Scientific American*, August 1991.
- Bothun, Gregory D., “Beyond the Hubble Sequence,” *Sky and Telescope*, May, 2000.
- Chiappini, Cristina, “Tracing the Milky Way’s History,” *Sky and Telescope*, October, 2004.
- Dressler, Alan, “Observing Galaxies through Time,” *Sky and Telescope*, August 1991.
- Keel, William C., “Crashing Galaxies, Cosmic Fireworks,” *Sky and Telescope*, January, 1989.
- Miller, M. Coleman, Reynolds, Christopher S., and Krishnamurthi, Anita, “Supermassive Black Holes: Shaping Their Surroundings,” *Sky and Telescope*, April, 2005.
- Schroeder, Michael C., and Comins, Neil F., “Galactic Collisions on Your Computer,” *Astronomy*, December, 1988.
- Silk, Joseph, *The Big Bang*, Third Edition, W. H. Freeman and Company, New York, 2001. *Space Telescope Science Institute*, <http://www.stsci.edu>.
- Toomre, Alar, and Toomre, Juri, “Violent Tides between Galaxies,” *Scientific American*, December, 1973.
- van den Bergh, Sidney, and Hesser, James E., “How the Milky Way Formed,” *Scientific American*, January, 1993.

Technical

- Arp, H. C., “Atlas of Peculiar Galaxies,” *The Astrophysical Journal Supplement*, 14, 1, 1966.
- Barnes, Joshua E., and Hernquist, Lars, “Computer Models of Colliding Galaxies,” *Physics Today*, March, 1993.
- Bender, Ralf, et al., “HST STIS Spectroscopy of the Triple Nucleus of M31: Two Nested Disks in Keplerian Rotation around a Supermassive Black Hole,” *The Astrophysical Journal*, 631, 280, 2005.
- Bertin, Giuseppe, *Dynamics of Galaxies*, Cambridge University Press, Cambridge, 2000.
- Binney, James, “Warps,” *Annual Review of Astronomy and Astrophysics*, 30, 51, 1992.
- Binney, James, and Merrifield, Michael, *Galactic Astronomy*, Princeton University Press, Princeton, NJ, 1998.
- Binney, James, and Tremaine, Scott, *Galactic Dynamics*, Princeton University Press, Princeton, NJ, 1987.
- Burkert, A., Truran, J. W., and Hensler, G., “The Collapse of Our Galaxy and the Formation of the Galactic Disk,” *The Astrophysical Journal*, 391, 651, 1992.

- Butcher, Harvey, and Oemler, Augustus Jr., "The Evolution of Galaxies in Clusters. I. ISIT Photometry of Cl 0024+1654 and 3C 295," *The Astrophysical Journal*, 219, 18, 1978.
- Chabrier, Gilles, "Galactic Stellar and Substellar Initial Mass Function," *Publications of the Astronomical Society of the Pacific*, 115, 763, 2003.
- Dressler, Alan, "Galaxy Morphology in Rich Clusters: Implications for the Formation and Evolution of Galaxies," *The Astrophysical Journal*, 236, 351, 1980.
- Eggen, O. J., Lynden-Bell, D., and Sandage, A. R., "Evidence from the Motions of Old Stars That the Galaxy Collapsed," *The Astrophysical Journal*, 136, 748, 1962.
- Elmegreen, Bruce G., and Scalo, John, "The Effects of Star Formation History on the Inferred Stellar Initial Mass Function," *The Astrophysical Journal*, 636, 149, 2006.
- Elmegreen, Debra Meloy, *Galaxies and Galactic Structure*, Prentice-Hall, Upper Saddle River, NJ, 1998.
- Kormendy, John, and Kennicutt, Robert C., Jr., "Secular Evolution and the Formation of Pseudobulges in Disk Galaxies," *Annual Review of Astronomy and Astrophysics*, 42, 603, 2004.
- Larson, Richard B., "Galaxy Building," *Publications of the Astronomical Society of the Pacific*, 102, 709, 1990.
- Mo, H. J., Mao, Shude, and White, Simon D. M., "The Formation of Galactic Discs," *The Monthly Notices of the Royal Astronomical Society*, 295, 319, 1998.
- Silk, Joseph, "Formation and Evolution of Disk Galaxies," *Astrophysics and Space Science*, 284, 663, 2003.
- Spinrad, Hyron, *Galaxy Formation and Evolution*, Praxis Publishing Ltd., Chichester, 2005.
- Toomre, Alar, and Toomre, Juri, "Galactic Bridges and Tails," *The Astrophysical Journal*, 178, 623, 1972.

PROBLEMS

- 26.1** (a) The mass density of stars in the neighborhood of the Sun is approximately $0.05 \text{ M}_{\odot} \text{ pc}^{-3}$ (see Section 24.2). Assuming that the mass density is constant and that all of the stars are main-sequence M stars, estimate the fraction of the Galactic disk's volume that is occupied by stars.
- (b) Suppose that an intruder star (a main-sequence M star) travels perpendicularly through the Galactic disk. What are the odds of the intruder colliding with another star during its passage through the disk? Take the thickness of the disk to be approximately 1 kpc.
- 26.2** Write a general expression for the force of dynamical friction as

$$f_d \simeq C (GM)^a (v_M)^b \rho^c,$$

where C is dimensionless, and a , b , and c are constants. Set up a system of three linear equations for a , b , and c such that f_d has units of force. Solve the system and show that Eq. (26.1) is obtained.

- 26.3** Use the age of the Milky Way's oldest globular clusters from Section 24.2 and the orbital velocity of the local standard of rest to estimate the greatest distance from which globular

clusters could have spiraled into the nucleus due to dynamical friction. Use $5 \times 10^6 M_\odot$ for the cluster's mass. Compare your answer with the size of the Milky Way's central bulge.

- 26.4** (a) The Large Magellanic Cloud, which has a mass of about $2 \times 10^{10} M_\odot$, orbits the Milky Way at a distance of 51 kpc. Assuming that the Milky Way's dark-matter halo and flat rotation curve extend out to the LMC, estimate how much time it will take for the LMC to spiral into the Milky Way (take $C = 23$). (Note that the Small Magellanic Cloud may be gravitationally bound to the LMC, but its mass is an order of magnitude smaller; therefore, the SMC will be neglected in this problem.)
- (b) Compare this rough estimate with the value quoted in the text for the merger time of the LMC (which was based on a detailed computer simulation of the LMC's dynamical interaction with our Galaxy).
- 26.5** Use Eq. (18.10) and the information in Problem 26.4 to estimate the tidal radius of the Large Magellanic Cloud as it orbits the Milky Way. Take the mass of the Milky Way's dark-matter halo to be $5.4 \times 10^{11} M_\odot$ from Table 24.1. How does your answer compare with the size of the LMC, which has an angular diameter of $460'$?
- 26.6** The Small Magellanic Cloud is 60 kpc away and has one-tenth the mass of the LMC (see Problem 26.4). The SMC has an angular diameter of $150'$, and its angular separation from the LMC is 21° .
- (a) What is the distance between the LMC and the SMC?
- (b) Ignoring the influence of the Milky Way, estimate the present tidal radii of both the LMC and the SMC due to the other.
- (c) The Magellanic Clouds are in eccentric orbits around one another, and during the last few hundred million years they have been moving apart with an average speed of about 110 km s^{-1} . How recently did the LMC extend beyond its tidal radius? How recently did the SMC extend beyond its tidal radius? Assuming that the Magellanic Stream was formed by tidal stripping when the LMC and SMC were close to each other, when would you estimate that the Magellanic Stream was formed? (Note: Following this mutual tidal stripping, the gas was pulled from the LMC/SMC system by our Galaxy.)
- 26.7** The gaseous ring orbiting the central galaxies of the M96 group has an orbital period of 4.1×10^9 years.
- (a) What is the combined mass (in $M_\odot$) of the two galaxies?
- (b) The combined visual luminosity of the central galaxies M105 and NGC 3384 is $L_V = 2.37 \times 10^{10} L_\odot$ (from Schneider, *Ap. J.*, 343, 94, 1989). What is the mass-to-light ratio for the pair?
- 26.8** Suppose that the original density distribution of the proto-Galactic nebula envisioned by Eggen, Lynden-Bell, and Sandage had a radial functional dependence similar to the dark-matter halo [assume the form of Eq. (24.51) with $r \gg a$]. Beginning directly with the radial equation of motion,

$$\frac{d^2 r}{dt^2} = -\frac{GM_r}{r^2},$$

show that the free-fall time would be proportional to radius.

Hints: Multiply both sides of the equation by $v = dr/dt$ and use the relation

$$v \left(\frac{dv}{dt} \right) = \frac{1}{2} \frac{dv^2}{dt}.$$

You may also find the following definite integral helpful:

$$\int_0^1 \frac{du}{\sqrt{\ln\left(\frac{1}{u}\right)}} = \sqrt{\pi}.$$

- 26.9** (a) Assume an initial mass function of the form given in Eq. (26.5). If $x = 1.8$, calculate the ratio of the number of stars that are formed in the mass range between $2 M_\odot$ and $3 M_\odot$ to those formed with masses between $10 M_\odot$ and $11 M_\odot$.
- (b) Beginning with the initial mass function and using the mass–luminosity relation for main-sequence stars, Eq. (24.12) (see also Fig. 7.7 and Problem 10.27), derive an expression for the number of main-sequence stars formed per unit luminosity interval, dN/dL .
- (c) If $x = 0.8$ and $\alpha = 4$, calculate the ratio of the number of stars that are formed with main-sequence luminosities between $2 L_\odot$ and $3 L_\odot$ to the number of those formed with luminosities between $10 L_\odot$ and $11 L_\odot$.
- (d) Compare your answers in parts (a) and (c), and explain the results in terms of the physical characteristics of stars along the main sequence.
- 26.10** Use the value for the star formation rate given on page 1019 to estimate the number of stars currently being formed per year in the Milky Way Galaxy. Compare your answer to the value quoted in the text. Clearly describe the assumptions (if any) that you made in arriving at your estimate.
- 26.11** (a) Using the Jeans mass (Eq. 12.14), estimate the upper mass limit of galaxies that are in virial equilibrium. Use the virial temperature and the number density determined in Example 26.2.2.
- (b) Assuming that the least massive galaxies could have formed out of a gas with an initial temperature near the ionization temperature of hydrogen, estimate the lower mass cutoff.
- (c) By using the Jeans length (Eq. 12.16), estimate the radius of the largest galaxies that are in virial equilibrium and compare your answer to the approximate size of our Galaxy.
- 26.12** (a) By equating the cooling timescale to the free-fall timescale, show that the maximum mass of a protogalactic nebula is given by

$$M = \frac{25}{32} \frac{\Lambda^2}{G^3 \mu^4 m_H^4 R}.$$

- (b) Estimate the maximum mass of a protogalactic nebula that can undergo a free-fall collapse if $R = 60$ kpc. Assume that $\Lambda \simeq 10^{-37} \text{ W m}^3$.
- 26.13** (a) M87 is a giant elliptical galaxy near the center of the Virgo cluster of galaxies. It is estimated that its mass is approximately $3 \times 10^{13} M_\odot$ within a radius of 300 kpc. How long would it take for a star near the outer edge of the galaxy to orbit the center once? Compare your answer to the approximate age of our Galaxy.
- (b) Based on your result from part (a) and assuming that M87 has been capturing smaller satellite galaxies up until the present time, would you expect the outer portions of M87 to be in virial equilibrium? Why or why not?
- 26.14** (a) From the data given in Problem 26.13, calculate the velocity of a star near the outer edge of the galaxy that may be following a circular orbit.
- (b) Assuming that the velocity you found in part (a) is characteristic of the velocity dispersion of material near the outer edge of the galaxy, estimate the virial temperature of M87.

- (c) What is the characteristic cooling time for M87? Use $\Lambda \simeq 10^{-37} \text{ W m}^3$.
 - (d) If the mass in the outer portions of M87 were to undergo a radial free-fall collapse, estimate the free-fall timescale for such a collapse and compare your answer with the cooling timescale.
- 26.15** Estimate the scale height associated with a Galactic disk having a temperature of 10^4 K . With which component of our Galaxy might such a disk correspond today?
- 26.16** It has been pointed out many times that given the current rate of star formation, the disk of our Galaxy will be depleted of interstellar gas in the relatively near future. From the data given in Table 24.1 and Section 26.2, estimate how much longer the Milky Way can effectively create new stars without an appreciable infall of new material. How might the period of star formation be prolonged?
- 26.17** (a) From the data in Table 25.4, estimate the escape speed from a dwarf elliptical galaxy.
 (b) Compare your answer in part (a) with the speed of the ejecta from a typical Type II supernova.
 (c) Compare your answer in part (a) with the thermal velocities of hydrogen ions in a gas having a temperature of 10^6 K .
 (d) How might you explain the evolution of ancient, gas-poor systems such as dE's and dSph's?

COMPUTER PROBLEMS

- 26.18** Figure 25.17 shows the Whirlpool galaxy, M51, and its companion, NGC 5195. Use the program `Galaxy`, described in Appendix M, to reproduce the appearance of this system. Use 10 rings of 50 stars each, and let the intruder galaxy have one-fourth the mass of the target galaxy. Use $(x, y, z) = (30, -30, 0)$ for the initial position of the intruder galaxy, and use $(v_x, v_y, v_z) = (0, 0.34, 0.34)$ for its velocity (in the units used by the program). Follow the motion for 648 million years (540 time steps).
- (a) When does the calculated appearance of the target and host galaxies resemble Fig. 25.17? Indicate the simulation times.
 - (b) What happens to the system near the end of this time period?
 - (c) Rerun the program, changing the initial x coordinate from 30 to -30 (and leaving everything else the same). Describe the difference in the outcomes.
- 26.19** `Galaxy` was used in Problem 26.18 to reproduce some the features of the Whirlpool galaxy, shown in Fig. 25.17. Run the program again, using the same values of the mass, initial position, and initial velocity of the intruder galaxy, but use only *two* rings with 24 stars each.
- (a) Compare the initial speed of the intruder galaxy with the orbital speeds of the nearby stars. Describe (in words) how the spiral pattern forms. Follow the motion for 360 million years (300 time steps).
 - (b) Rerun the program, changing the initial x coordinate from 30 to -30 (and leaving everything else the same). Describe and physically explain the differences between this and the behavior of the stars in part (a).
- 26.20** Figure 26.6 shows the Cartwheel galaxy, the result of a high-speed, head-on collision. Use the program `Galaxy`, described in Appendix M, to investigate how a nearly head-on collision may

have produced this ring galaxy. Use 10 rings of 24 stars each, and assume that the galaxies involved have equal masses.

- (a) Try an exact head-on collision first. Let $(x, y, z) = (0, 0, 35)$ for the initial position of the intruder galaxy, and let $(v_x, v_y, v_z) = (0, 0, -1)$ for its velocity (in the units used by the program). Follow the motion for 78 million years (65 time steps), and describe what happens. Could such a process be responsible for Fig. 26.6?
- (b) Explain how the force of gravity, always acting inward, could have caused the rings of stars to *expand*.
- (c) Investigate what happens if the initial value of the x coordinate varies between 0 and 10 (with everything else unchanged). Which values (roughly) do the best job of reproducing the appearance of a ring galaxy with the least disruption of the target galaxy?

The Structure of the Universe

27.1 *The Extragalactic Distance Scale*

27.2 *The Expansion of the Universe*

27.3 *Clusters of Galaxies*

27.1 ■ THE EXTRAGALACTIC DISTANCE SCALE

The drama of the heavens unfolds on a two-dimensional stage. Observations readily supply the two coordinates, right ascension and declination, that specify an object's position on the celestial sphere. However, the object's distance is not so easily obtained. Astronomers today use many methods to estimate the distances of remote galaxies. In this section we will introduce a variety of techniques; some depend on the properties of individual stars, some use other objects (globular clusters, planetary nebulae, supernovae, HII regions), and some rely on the statistical properties of whole galaxies and galaxy clusters. The third dimension of distance is doubly significant because, as we saw at the end of Section 26.2, when astronomers peer deeper and deeper into space, they are looking back in time at increasingly ancient light. It is vital to understand not only a galaxy's position in space, but its depth in time as well.

Unveiling the Third Dimension

The techniques used to determine distances within the solar neighborhood were described in Section 3.1. In 1761, the method of trigonometric parallax was used to measure the distance to Venus, thereby calibrating the size of Kepler's Solar System. When Friedrich Wilhelm Bessel measured the subtle annual shift in the position of the star 61 Cygni in 1862, he combined the parallax method with his knowledge of the true size of Earth's orbit to discover that 61 Cygni is 650,000 times farther away than the Sun. Today, the surveyor's method of trigonometric parallax can reach out to a kiloparsec or so.¹ We have also seen how the moving cluster method (Section 24.3) made it possible to determine the distance of the Hyades star cluster. From there, the technique of main-sequence fitting (Sections 13.3 and 24.3) can be used to find the distances to open clusters out to about 7 kpc by comparing their main sequences on an H–R diagram with that of the Hyades cluster. The repeated application

¹The distances to the nearer planets can be determined within a few kilometers by measuring the round-trip time taken by radar waves sent from Earth and reflected back from the planet's surface. However, because the strength of the reflected wave diminishes rapidly (as $1/r^4$), this method is limited to within about a light-hour, or 10^9 km.

of a variety of methods using this pattern of calibration and measurement constitutes the steps of the **extragalactic distance scale**, also referred to as the **cosmological distance ladder**.

The Wilson–Bappu Effect

Spectroscopic parallax (discussed in Section 8.2) can in principle provide reliable distances to remote stars out to about 7 Mpc away, although in practice it is employed within a few hundred kiloparsecs, far enough to reach the Magellanic Clouds. Some stars have specific features in their spectra that allow their absolute magnitudes, and hence their distances, to be calculated. For example, the K absorption line of calcium can be quite broad, reaching maximum strength at spectral type K0 (Section 8.1). In late-type stars with chromospheres (types G, K, and M), a narrow emission line is seen, centered on the wide K absorption line. The width of this emission line is strongly correlated with a star’s absolute visual magnitude. This **Wilson–Bappu effect**, discovered in 1956 by Olin C. Wilson, Jr. (1909–1994) and M. K. Vainu Bappu (1927–1982), is valid for stars over a range of 15 magnitudes.

The Cepheid Distance Scale

For more distant objects, astronomers turn to the period–luminosity relation for Cepheids, discovered by Henrietta Leavitt (Section 14.1). Before this relation could be used, it had to be calibrated by finding the distance to a classical Cepheid. Unfortunately, the nearest one, Polaris, was too far away (≈ 200 pc) for trigonometric parallax to be useful. In 1913, however, Ejnar Hertzsprung used the Sun’s motion of 16.5 km s^{-1} with respect to the local standard of rest (Section 24.3) to provide a longer baseline for parallax measurements. It was this technique of **secular parallax** that enabled him to determine the average distance to a classical Cepheid with a period of 6.6 days. Hertzsprung then used this information to calibrate the period–luminosity relation; Harlow Shapley soon carried out a similar procedure. Since then, the distances to more Cepheids have been measured by parallax and other methods, and the period–luminosity relation has become well established. As mentioned in Section 14.1, astronomers today use a period–luminosity–color relation such as

$$M_{(V)} = -3.53 \log_{10} P_d - 2.13 + 2.13(B - V), \quad (27.1)$$

in the V band to account for the finite width of the instability strip on the H–R diagram [you should compare this with the period–luminosity relation, Eq. (14.1) in the V band]. Here, P_d is the pulsation period in units of days, and $B - V$ is the color index; for classical Cepheids, $B - V \approx 0.4$ to 1.1 . After the star’s absolute magnitude has been calculated, it can be combined with the star’s apparent magnitude to give its distance modulus; Eq. (24.1) then provides the star’s distance *if the extinction (in magnitudes) is known*.

Cepheid variable stars immediately proved their value as stellar yardsticks. It was in 1917 that Shapley measured the distances to Population II Cepheids in globular clusters, thereby determining his estimates of 100 kpc for the diameter of the Galaxy and 15 kpc for

the distance of the Sun from its center.² Then, in 1923, Edwin Hubble discovered several Cepheids in the Andromeda galaxy and announced that it was 285 kpc away. (The modern value is 770 kpc; see Section 25.1.) It was Hubble's series of observations that established M31 as an external galaxy, not a smaller nebula within the borders of the Milky Way.

It should be noted that the existence of a correlation between the period of a Cepheid's oscillations and its absolute magnitude meant that these stars could be used as standard candles to determine distances many years *before* the physical processes that cause the pulsations were understood. This involves some risk, and in this case there is a cautionary tale worth telling. At the time, it was not known that *three* types of pulsating stars were being used to determine the size of the Galaxy and the distance to M31. Furthermore, although the existence of interstellar dust clouds had been established by Edward Barnard (1857–1923) in the early years of the twentieth century, the existence of diffuse interstellar dust and gas capable of extinguishing starlight had yet to be demonstrated. Leavitt's variables in the Magellanic Clouds were classical Cepheids (Population I stars), while those observed in the globular clusters by Shapley were W Virginis and RR Lyrae stars (both Population II).³ The W Virginis stars are about 1.5 magnitudes fainter than classical Cepheids of the same period, which corresponds to their being less luminous by a factor of 4. When Hertzsprung and later Shapley calibrated the period–luminosity relation, they used nearby classical Cepheids but neglected the effect of extinction. Neither realized that dust in the Galactic disk dimmed these stars by (coincidentally) about 1.5 magnitudes.

Thus by chance, the resulting period–luminosity relation was just about right when used with W Virginis stars. However, it would give an underluminous result with classical Cepheids. When Edwin Hubble used this calibration to find the distances to M31 and other galaxies (which were observed out of the obscuring plane of the Galaxy), the apparent magnitudes of his Cepheids were correct but his estimates of the absolute magnitudes were too large. Consequently, their distance moduli ($m - M$) and the distances themselves were underestimated. The stars were thought to be dimmer and closer, rather than brighter and farther away, and so their parent galaxies were determined to be at roughly *half* of their actual distances and *half* their actual sizes.

Harlow Shapley fared no better with his observations of the (unknown to him) W Virginis stars in the globular clusters. Although he unwittingly used the right variety of star for the calibration of the period–luminosity relation, Shapley's observations of W Virginis stars were made in systems near the Galactic plane. Extinction due to dust within the disk dimmed the starlight and increased the stars' apparent magnitudes. Because the importance of extinction was not yet recognized, Shapley incorrectly attributed his stars' faintness to their remoteness. As a result, his distances to the globular clusters were too great, as was the size he calculated for the Galaxy.

Something was obviously wrong. It appeared that the Milky Way was much larger than any of the other galaxies surveyed. This is just the sort of anthropomorphic preeminence that makes astronomers uncomfortable. Robert Trumpler's (1886–1956) proof of interstellar

²Recall that Shapley's values for the dimensions of the Milky Way have since been revised substantially, an ongoing process described in Section 24.2.

³Shapley used clusters containing both W Virginis and RR Lyrae stars to calibrate a period–luminosity relation for the RR Lyraes. He then used observations of RR Lyrae stars in those clusters lacking the brighter W Virginis variables, and so the error was propagated.

extinction in 1930 remedied part of the problem. The rest of the puzzle was solved in 1952 when Walter Baade announced that there are two types of Cepheids: the classical Cepheids and the intrinsically fainter W Virginis stars.⁴ With these corrections, the measured distances and sizes of the other galaxies doubled and the Milky Way was reduced to being a bit smaller than M31.

In the early 1990s, the calibration of the period–luminosity relation was carried out using data obtained by the Hipparcos mission, described in Section 3.1. Hipparcos astronomers measured the parallax angles of 273 Cepheids and used the resulting distances to derive the period–luminosity relation quoted in Eq. (14.1). This required the cooperation of amateur astronomers, whose careful observations of light curves made it possible to measure the brightness of these variables at the appropriate phases. This was the first period–luminosity relation to be obtained from a direct distance measurement, and it is interesting that the result was nearly identical to an expression obtained in 1968 by American astronomer Allan Sandage and his Swiss colleague Gustav Tammann. Although further adjustments are sure to follow, Cepheids certainly provide a firm foundation for measuring the distances to other galaxies.

Interstellar extinction is still the largest source of error when Cepheids are used as standard candles, and there may also be a weak dependence on metallicity. The extinction problem can be reduced by observing these stars at infrared wavelengths, since infrared light more readily penetrates dusty regions (recall Fig. 14.6). However, because Cepheids are about three magnitudes dimmer in the infrared, astronomers have not stopped searching for them at visible wavelengths.

The Cepheid-based distances to galaxies within the Virgo cluster of galaxies ranges from 15 Mpc to 25 Mpc. The variation in these values is primarily due to the large spatial extent of the Virgo cluster along the line of sight, and to the fact that the spiral galaxies in which the Cepheids are observed are found primarily on the outskirts of the cluster and not in its core. To date, the most remote classical Cepheids known are 29 Mpc distant and reside in NGC 3370, a spiral galaxy in the direction of the constellation Leo.⁵ Overall, the uncertainty in the distances obtained using classical Cepheids as standard candles ranges from 7% for the LMC to perhaps 15% for more distant galaxies.

Supernovae as Distance Indicators

Supernovae can be used in several ways to measure extragalactic distances. Suppose the angular extent, $\theta(t)$, of a supernova's photosphere is observed. The angular velocity of the expanding gases, $\omega = \Delta\theta/\Delta t$, can be found by comparing two observations separated by a time Δt . If d is the distance to the supernova, then the transverse velocity of the expanding photosphere is $v_\theta = \omega d$. Assuming that the expansion is spherically symmetric, this transverse velocity should be equal to the ejecta's radial velocity, v_{ej} , obtained from the supernova's Doppler-shifted spectral lines. Then the distance to the supernova is

$$d = \frac{v_{ej}}{\omega}.$$

⁴Baade had described Population I and II stars in 1944 and had wondered at the time whether there might be two corresponding varieties of Cepheids.

⁵See Riess et al. (2005) for details of the observations of these remote Cepheids.

Most supernovae are too distant for this method to be employed, so another strategy is adopted. It assumes that the expanding shell of hot gases radiates as a blackbody (to a first approximation). Then the supernova's luminosity is given by the Stefan–Boltzmann law, Eq. (3.17),

$$L = 4\pi R^2(t)\sigma T_e^4,$$

where $R(t)$ is the radius of the expanding photosphere and t is the age of the supernova. If we assume that the ejecta's radial velocity has remained nearly constant, then $R(t) = v_{ej}t$. The effective temperature of the photosphere comes from the characteristics of its blackbody spectrum. Once the luminosity is found, it can be converted to an absolute magnitude (via Eq. 3.8) and then used to find the distance to the supernova by comparing it with the observed apparent magnitude. Of course, the photosphere of the expanding shell of a supernova is neither perfectly spherical nor a perfect blackbody. Difficulties with accurate values for interstellar extinction plague both methods, but the problem is more acute for core-collapse supernovae (Types Ib, Ic, and II), which are found near sites of recent star formation. Typical uncertainties in the distances obtained range from 15% (for M101) to 25% (for the Virgo cluster of galaxies).

Example 27.1.1. Twenty-five days after maximum light, the spectrum of an “average” Type Ia supernova is that of a blackbody with an effective temperature of 6000 ± 1000 K. The speed of the shell's photosphere, obtained from its Doppler-shifted absorption lines, is 9500 ± 500 km s^{−1}. If the rise time to maximum light is 17 ± 3 days, then when the average Type Ia supernova is $t = 42$ days old, its luminosity is

$$L \approx 4\pi (v_{ej}t)^2 \sigma T_e^4 = 1.10 \times 10^{36} \text{ W}. \quad (27.2)$$

From Eq. (3.8) its absolute bolometric magnitude at this time is

$$M_{bol} = M_{Sun} - 2.5 \log_{10}(L/L_{\odot}) \approx -18.9.$$

Type Ia Light Curves

The most important way of using supernovae to measure distance takes advantage of the similarity of Type Ia light curves. As discussed in Section 18.5, these supernovae have blue and visual absolute magnitudes of $\langle M_B \rangle \simeq \langle M_V \rangle \simeq -19.3 \pm 0.3$ at maximum light. If the peak magnitude of a Type Ia can be determined, its distance may then be determined.

Much effort has been devoted to improving our understanding of Type Ia light curves; recall Fig. 18.21. Fortunately there is a well-defined inverse correlation between a Type Ia's maximum brightness and the rate of decline of its light curve, and astronomers have learned how to use this information to more precisely determine the supernova's intrinsic peak luminosity.

In practice, a supernova is observed over time at several wavelengths. The **multicolor light curve shapes (MCLS) method** then compares the shape of the light curve to a family of parameterized template curves that allows the absolute magnitude of the supernova at maximum brightness to be determined, even if the supernova was not caught at its peak

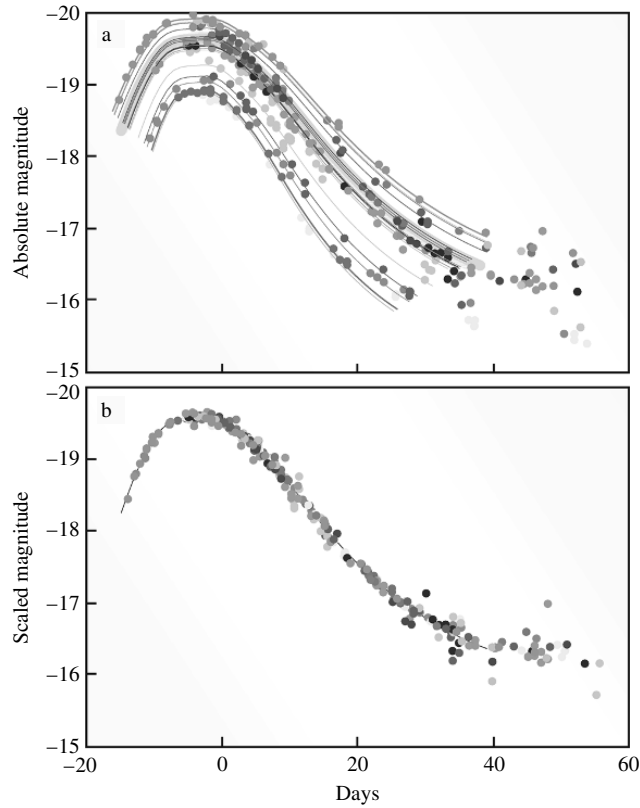


FIGURE 27.1 Low-redshift Type Ia template light curves. (a) The light curves of several Type Ia supernovae, as measured. (b) The light curves after applying the time scale stretch factor. The blue absolute magnitude is displayed on the vertical axis. (Figure adapted from Perlmutter, *Physics Today*, 56, No. 4, 53, 2003.)

brilliance. The MLCS method also allows the reddening and dimming effect of interstellar dust to be detected and removed.

Another approach is the **stretch method**, which fits the B and V magnitude light curves with a single template light curve that has been stretched (or compressed) in time (see Fig. 27.1). The peak magnitude is then determined by the stretch factor. These techniques allow astronomers to use Type Ia supernovae to determine distances with an uncertainty approaching just 5%, corresponding to an uncertainty in the distance modulus, $m - M$, of 0.1 magnitude.

Example 27.1.2. The Type Ia supernova SN 1963p in the galaxy NGC 1084 had an apparent blue magnitude of $B = 14.0$ at peak brilliance. Then, with an extinction of 0.49 mag to that galaxy, the distance to the supernova is approximately

$$d = 10^{(m-M-A+5)/5} = 36.5 \text{ Mpc.}$$

Because Type Ia supernovae are about 13.3 magnitudes brighter than the brightest Cepheid variables (-19.3 compared to -6), this method is capable of reaching out more than 500 times farther than with Cepheids, providing estimates to truly cosmological distances exceeding 1000 Mpc.⁶ Sophisticated supernova search programs are carried out that more than compensate for the unlikely probability of detecting an explosion in a specific galaxy by scanning large numbers of galaxies in a wide field of view. Near the end of the twentieth century, two teams of astronomers made careful observations of high-redshift Type Ia supernovae and discovered that the expansion of the universe is accelerating, a story that will be recounted in detail in Section 29.4. The Supernova/Acceleration Probe (SNAP) spacecraft is currently being designed and may be launched in 2010. Its mission will be to observe thousands of distant Type Ia supernovae and to map several percent of the sky for gravitational lenses every year.

Using Novae in Distance Determinations

From the sizes of their expanding photospheres, novae can be used in the same manner as supernovae to derive distances. In addition, although there is a wide variation in the absolute magnitudes of novae at peak light, there is a relation between a nova's maximum visual magnitude, $M_V^{\max}$, and the time it takes for its visible light to decline by two magnitudes.⁷ Consequently, novae can also serve as standard candles.

The physical reason why this relationship holds for novae is that more massive, smaller white dwarfs produce greater compression and heating of the accreted gases on their surfaces, and so a runaway thermonuclear reaction may be initiated with a smaller accumulated mass; see Section 18.5. The less massive surface layers are more readily ejected, and so the nova declines more rapidly. Writing the average rate of decline over the first 2 mag as $\dot{m}$ (in mag d⁻¹), this relation can be expressed as

$$M_V^{\max} = -9.96 - 2.31 \log_{10} \dot{m} \quad (27.3)$$

for Galactic novae, with an uncertainty of about ± 0.4 mag. After fading by two magnitudes, the brightest novae are about as luminous as the brightest Cepheids, so these two methods span the same region of space (to about 20 Mpc, just past the Virgo cluster of galaxies).

Secondary Distance Indicators

With the exception of techniques associated with unpredictable supernovae, in order for astronomers to reach farther into space and to more remote galaxies, *secondary* methods must be used to measure distance. Unlike the *primary* methods already described, these secondary distance indicators require a galaxy with an established distance for their calibration. One way of seeing farther involves using the brightest objects in a galaxy. For example, the three brightest giant H II regions in a galaxy have been used to provide a standard candle. These regions, which may contain up to $10^9 M_\odot$ of ionized hydrogen, are visible at great distances. Measurements of the angular sizes of the H II regions and the apparent magnitude of the galaxy can be compared with similar measurements for other

⁶Type II supernovae are dimmer than Type Ia's by about 2 mag and, therefore, at best can be seen only about 40% as far away.

⁷There are several variants of this relation that use the time for the nova to decline by 2 mag, 3 mag, etc.

galaxies with known distances. This makes it possible to calculate the H II regions' linear sizes (in parsecs), along with the galaxy's absolute magnitude. However, since the diameter of an H II region is difficult to define unambiguously, this procedure is relatively insensitive to distance and must be used carefully.

The brightest red supergiants seem to have about the same absolute visual magnitude in all galaxies, as shown by careful studies of the brightest galactic stars carried out by Roberta Humphreys. Apparently, mass loss reduces the brightest red supergiants to about the same maximum mass, which results in their having about the same luminosity. Sampling a number of galaxies revealed the average visual magnitude of the three brightest red stars to be $M_V = -8.0$. Because individual stars must be resolved in order for this method to be used, its range is the same as for spectroscopic parallax, about 7 Mpc.

The Globular Cluster Luminosity Function

The limited sampling inherent in these methods ("the three brightest ...") can lead to errors. It is statistically more secure to take as complete an inventory as possible of some class of objects associated with a galaxy, and then describe how the objects vary with magnitude. For example, Fig. 27.2 shows the **globular cluster luminosity function**, $\phi(M_B)$, for the globular clusters around four giant elliptical galaxies in the Virgo cluster.⁸ In the figure, $\phi(M_B) dM_B$ is the number of globular clusters having absolute blue magnitudes between M_B and $M_B + dM_B$. The distribution is well described by a Gaussian function (the solid

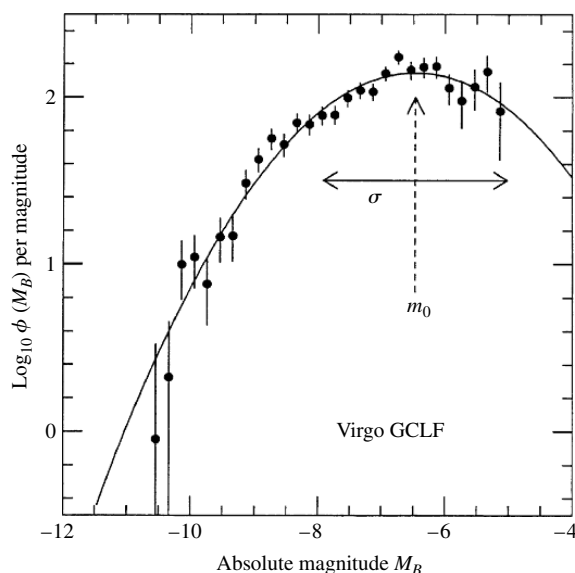


FIGURE 27.2 The luminosity function for the globular clusters around four giant elliptical galaxies in the Virgo cluster. About 2000 clusters brighter than $B = 26.2$ were used. A distance of 17 Mpc was adopted for the Virgo cluster. [Adapted from a figure courtesy of Jacoby (KPNO/NOAO), as provided by Harris (McMaster Univ.) for *Publ. Astron. Soc. Pac.*, 104, 599, 1992.]

⁸Luminosity functions were introduced in Section 25.4.

line in the figure) that has a well-defined peak at a *turnover magnitude* of $M_0 \simeq -6.5$. (Notice that the value of M_0 depends on the distance to the Virgo cluster used in calculating the absolute magnitudes.) The value of the turnover magnitude provides a standard candle that can be used to find the distance to the globular clusters surrounding another galaxy.

The procedure is to measure the luminosity function for the galaxy under investigation and to compare its apparent turnover magnitude, m_0 , with M_0 for the Virgo cluster. The best results are achieved with galaxies having large numbers of globular clusters, such as giant ellipticals; recall Fig. 25.16. It is desirable that the calculated $\phi(m_B)$ extend well past the turnover point, although there are (less desirable) ways of fitting just the brighter end of the luminosity function. Overall, this method should yield a value for a galaxy's distance modulus that is uncertain by only about 0.4 mag, corresponding to an uncertainty in distance of about 20%. Globular clusters are visible from vast distances, and in the future this technique may reach out beyond the Virgo cluster to 50 Mpc.

Unfortunately, it is not yet certain whether there is a *universal* globular cluster luminosity function that applies to all types of galaxies, although for nine galaxies (including M31 and the Milky Way), the average value is $M_0 = -6.6 \pm 0.26$. The physical basis for this apparent agreement is unclear. However, globular clusters are ancient collections of stars, and it is reasonable to suppose that the subsequent evolution of the galaxies to which they are attached may not have significantly affected the individual clusters.

The Planetary Nebula Luminosity Function

A similar statistical analysis can be performed for a galaxy's planetary nebulae. Figure 27.3 shows the **planetary nebula luminosity function (PNLF)** for planetary nebulae in the Leo I group of galaxies as a function of their absolute magnitudes at a wavelength of

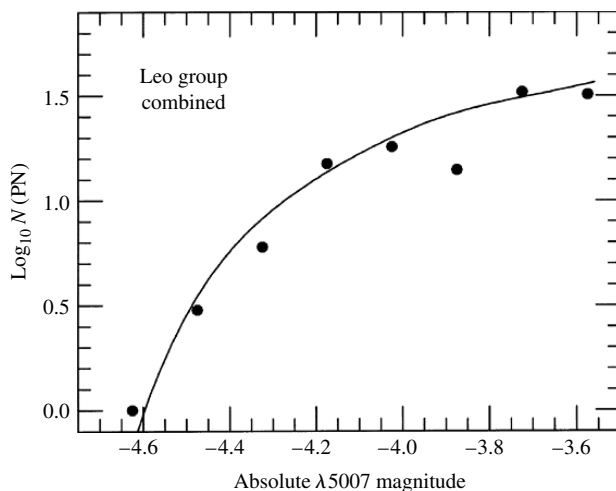


FIGURE 27.3 The planetary nebula luminosity function for the Leo I group of galaxies. (Figure adapted from Ciardullo, Jacoby, and Ford, *Ap. J.*, 344, 715, 1989.)

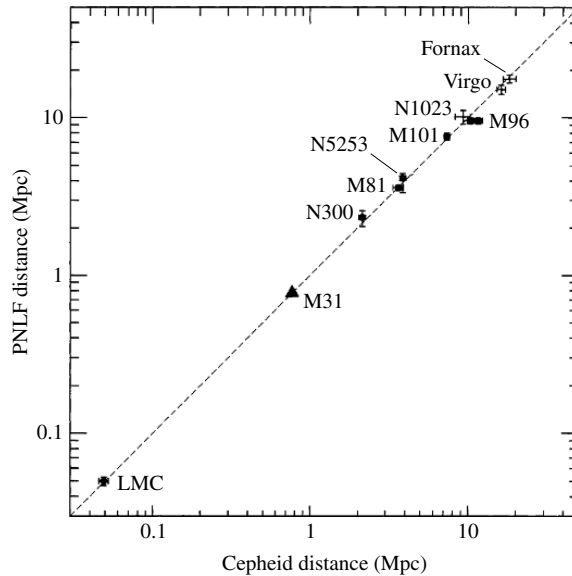


FIGURE 27.4 A comparison between distances obtained using the planetary nebula luminosity function and Cepheids. (Figure adapted from Jacoby, “Future Directions for the Planetary Nebula Luminosity Function” in *Harmonizing Cosmic Distance Scales in a Post-Hipparcos Era*, *ASP Conference Series*, 167, 175, 1999.)

500.7 nm.⁹ Using extragalactic planetary nebulae in this way is a reliable method of finding the distances to elliptical galaxies within about 20 Mpc. As a larger sample of galaxies was studied, the value determined for the cutoff decreased from the one depicted in Fig 27.3 to $M_{5007} = -4.53$. This can be adopted as a standard candle for the brightest planetary nebula. If the promise of this method is fulfilled, it should reach out to a distance of some 50 Mpc. Figure 27.4 shows the good agreement obtained between distances from the planetary nebula luminosity function and from Cepheid variables.

The Surface Brightness Fluctuation Method

Astronomers turn to the global properties of galaxies to probe even farther, up to 100 Mpc or more. One promising approach is to take advantage of how a detector such as a CCD camera records the appearance of a galaxy. Some pixels will record more stars than others due to spatial fluctuations in the galaxy’s surface brightness, but the overall appearance should become smoother with increasing distance. The results of a statistical analysis describe the magnitude of the pixel-to-pixel variation, and this is correlated with the galaxy’s distance. With the Hubble Space Telescope, the **surface brightness fluctuation method** could reach out to 125 Mpc, but it is usually applied more locally.

⁹This is a forbidden line of oxygen, [O III]. The observations are made through a filter with a bandwidth of about 3 nm; see Section 3.6.

The Tully–Fisher Relation

The Tully–Fisher relation for spiral galaxies (see Section 25.2) also provides a valuable tool for determining extragalactic distances. As discussed in Section 25.2, this is a relation between the luminosity of a spiral galaxy and its maximum rotation velocity. For the sake of completeness, we include an example of this important technique.

Example 27.1.3. For the galaxy M81, the quantity $W_R^i = 484 \text{ km s}^{-1}$. From Eq. (25.8), the infrared absolute magnitude is

$$M_H = -9.50(\log_{10} W_R^i - 2.50) - 21.67 = -23.43.$$

The apparent H magnitude of M81 is $H = 4.29$ (already corrected for extinction), and so according to the Tully–Fisher relation, the distance to M81 is

$$d = 10^{(H - M_H + 5)/5} = 3.50 \text{ Mpc}.$$

The appeal of the Tully–Fisher method of distance determination lies in its accuracy (typically ± 0.4 mag in the infrared, although for carefully selected targets the uncertainty may be reduced to ± 0.1 mag) and its great range (as far as 100 Mpc). Furthermore, nearby spirals whose distances have been accurately measured by Cepheids can be used for its calibration. The Tully–Fisher method has been applied to 161 spiral galaxies belonging to the Virgo Cluster. The results have produced a three-dimensional map of the cluster, and they show that the Virgo Cluster is extended along the line of sight by about four times its diameter on the sky. The Tully–Fisher method is also being used to study the Coma cluster of galaxies, about 90 Mpc away, and has been applied to distances well beyond that. It is one of the most widely used methods of measuring extragalactic distances.

The D – σ Relation

The analogous correlation for elliptical galaxies, the Faber–Jackson relation ($L \propto \sigma_r^4$), shows considerable scatter; recall Fig. 25.33. More recently, an improvement has appeared, the **D – σ relation**, that relates the velocity dispersion to the diameter, D , of an elliptical galaxy. More precisely, D is the galaxy’s angular diameter out to a surface brightness level of $20.75 \text{ B-mag arcsec}^{-2}$.¹⁰ Because the surface brightness is independent of the galaxy’s distance¹¹ (recall Sections 6.1 and 9.1), D is inversely proportional to the galaxy’s distance, d . If the galaxy were twice as far away, its angular diameter would be half as large. In this way, D provides a *standard ruler* rather than a standard candle. There is a tighter relationship between σ and D than there is between σ and L ; see Fig. 27.5. An empirical relation between $\log_{10} D$ and $\log_{10} \sigma$ for the galaxies in a cluster (all at about the same distance) is

$$\log_{10} D = 1.333 \log \sigma + C, \quad (27.4)$$

¹⁰Actually, this is just another representation of the *fundamental plane* discussed in Section 25.4, since it relates radius, surface brightness, and velocity dispersion.

¹¹This statement is technically true only if the galaxy is closer than a cosmological distance of roughly 500 Mpc, at which point spacetime curvature effects become significant (see Chapter 29).

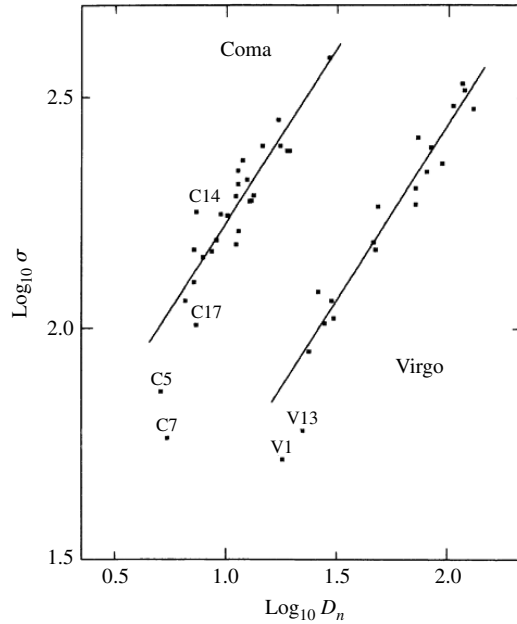


FIGURE 27.5 A logarithmic plot of diameters D (in arcseconds) and velocity dispersions σ (in km s^{-1}) for galaxies in the Virgo and Coma clusters. (Figure adapted from Dressler et al., *Ap. J.*, 313, 42, 1987.)

where the value of the constant C depends on the distance to the cluster of galaxies. Unfortunately, there are no bright elliptical galaxies available for the accurate calibration of this D – σ method by primary distance indicators such as Cepheids. However, because the slopes of the lines in the figure are very nearly the same, the vertical distance between the lines for two different clusters is

$$\log_{10} D_1 - \log_{10} D_2 = C_1 - C_2.$$

Then, because D is inversely proportional to a galaxy's distance, d , the *relative* distances between the two clusters can be found by using

$$\frac{d_2}{d_1} = \frac{D_1}{D_2} = 10^{C_1 - C_2}. \quad (27.5)$$

Example 27.1.4. In Eq. (27.4), the value of C for the Virgo cluster is -1.237 , and for the Coma cluster it is -1.967 . Thus the ratio of the distance of the Coma cluster of galaxies to that of the Virgo cluster is

$$\frac{d_{\text{Coma}}}{d_{\text{Virgo}}} = 10^{C_{\text{Virgo}} - C_{\text{Coma}}} = 5.37.$$

The Coma cluster is more than 5 times farther away than the Virgo cluster.

The $D-\sigma$ relation is a powerful tool for studying the distribution of groups of galaxies in space. Because of the inherent brightness of giant elliptical galaxies, this technique has the potential of exceeding the range of the Tully–Fisher relation.

The Brightest Galaxies in Clusters

Just as the brightest giant H II regions or the brightest red supergiants have been used to determine the distance to individual galaxies, the brightest galaxy in a cluster of galaxies can be used to obtain the cluster's distance. Figure 27.6 shows a composite galaxy luminosity function, similar to the ones given in Fig. 25.36. The sharp decline in the function at the bright end (small M) shows that the absolute magnitude of the brightest galaxy can be determined with some accuracy. The average value of the absolute visual magnitude for the brightest galaxy in ten nearby clusters is $M_V = -22.83 \pm 0.61$. This is 3.2 magnitudes brighter than a Type Ia supernova at peak brilliance, so in principle this method should reach four times farther than with supernovae, to over 4000 Mpc. The light received from such a distance is truly ancient, having traveled through space for 13 billion years.

The danger in using this method to look out to such great distances is that galaxies and clusters of galaxies both evolve, as discussed in Section 26.2. For instance, the brightest

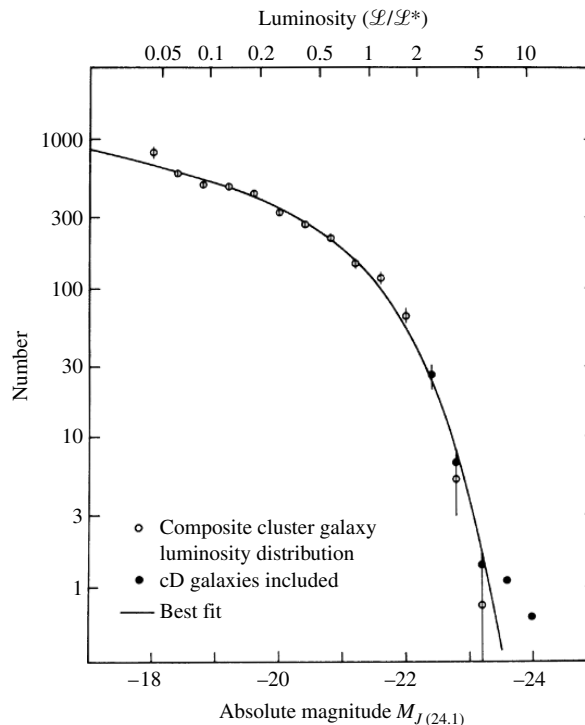


FIGURE 27.6 A composite luminosity function for the galaxies in several clusters. (Figure adapted from Schechter, *Ap. J.*, 203, 297, 1976.)

galaxy in a contemporary cluster may not have existed in its present form billions of years ago. As was mentioned in Section 26.1, the giant cD galaxies often found at the centers of rich clusters are probably the result of galactic mergers. Therefore, using a general galaxy luminosity function calibrated with nearby clusters may not be appropriate for very distant, and consequently younger-appearing, clusters.

A Summary of Distance Indicators

By now it should be clear that the extragalactic distance scale is not a simple ladder, with just one sequence of steps possible for measuring the greatest distances. Different astronomers use different techniques and choose different calibrations. There are variations with every method, and the relations described above are chosen only to be representative. As a summary, Table 27.1 lists the distance to the Virgo cluster of galaxies as determined by many of these methods. It also shows the uncertainty (in magnitudes) associated with each of these techniques. Overall, the distances listed are in good agreement, and we will adopt 16 Mpc as the distance to the Virgo cluster (see Problem 27.19). These methods are sufficiently accurate to determine the features of our corner of the universe, out to a distance of a few hundred Mpc.

TABLE 27.1 Distance Indicators. (Adapted from Jacoby et al., *Publ. Astron. Soc. Pac.*, 104, 599, 1992.)

Method	Uncertainty for Single Galaxy (mag)	Distance to Virgo Cluster (Mpc)	Range (Mpc)
Cepheids	0.16	15 – 25	29
Novae	0.4	21.1 ± 3.9	20
Planetary nebula luminosity function	0.3	15.4 ± 1.1	50
Globular cluster luminosity function	0.4	18.8 ± 3.8	50
Surface brightness fluctuations	0.3	15.9 ± 0.9	50
Tully–Fisher relation	0.4	15.8 ± 1.5	> 100
D – σ relation	0.5	16.8 ± 2.4	> 100
Type Ia supernovae	0.10	19.4 ± 5.0	> 1000

27.2 ■ THE EXPANSION OF THE UNIVERSE

In the first decade of the twentieth century, even before the extragalactic nature of spiral nebulae was understood, astronomers began making systematic observations of the radial velocities of galaxies by measuring their Doppler-shifted spectral lines. It was hoped that if the motions of these objects were found to be random, then the Sun's motion in the Milky Way should be related to the vector sum of the radial velocities of the nebulae.¹² It was V. M. Slipher, working at Lowell Observatory, who first discovered that this plan was doomed to failure. The velocities of the nebulae were *not* random. Instead, most of the spectra showed redshifted spectral lines. Slipher announced in 1914 that most of the 12 galaxies he had surveyed were rapidly moving *away* from Earth, although Andromeda's blueshifted spectrum showed it was approaching Earth at nearly 300 km s^{-1} . It was quickly realized that these galaxies not only were moving away from Earth but were moving away from each other as well. Astronomers began to talk about these galactic motions in terms of an expansion. Meanwhile, Slipher continued his measurements of radial velocities. By 1925 he had examined 40 galaxies and confirmed that spectra showing redshifted lines were much more common than those exhibiting blueshifts. Slipher concluded that nearly every galaxy he examined was rapidly receding from Earth.

Hubble's Law of Universal Expansion

1925 was also the year that Hubble discovered Cepheid variable stars in M31, thereby establishing that the Andromeda "nebula" was in fact an external galaxy. Hubble continued his search for Cepheids, determining the distances to 18 galaxies. He combined his results with Slipher's velocities and discovered that a galaxy's recessional velocity, v , was proportional to its distance, d . In 1929 Hubble presented his results in a paper, "A Relation between Distance and Radial Velocity among Extra-Galactic Nebulae," at a meeting of the National Academy of Sciences. This relation,

$$v = H_0 d, \quad (27.6)$$

is today called **Hubble's law**, and H_0 is the **Hubble constant**. Usually v is given in km s^{-1} and d in Mpc, so H_0 has units of $\text{km s}^{-1} \text{ Mpc}^{-1}$.

Hubble realized that he had discovered an exceptionally powerful way of finding distances to remote galaxies simply by measuring their redshifts. His early results show a straight line drawn through a scattering of points on a graph of velocity versus distance; see Fig. 27.7.

Hubble continued to compile distances and redshifts to strengthen this relationship, with much of the work done by his assistant, Milton Humason (1891–1972). Humason literally worked his way up to the top at the Mt. Wilson Observatory. He started out as a mule packer when the observatory was being constructed and served as a restaurant busboy, janitor, and night assistant at Mt. Wilson. After Humason got permission to do some observing on the

¹²This is basically the same procedure that was described in Section 24.3 for using nearby stars to find the Sun's peculiar motion relative to the local standard of rest; see Eqs. (24.30–24.32).

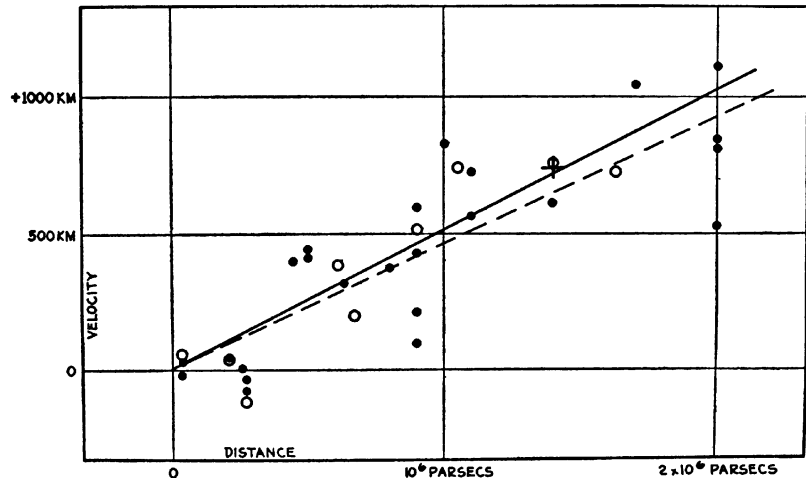


FIGURE 27.7 Hubble's 1936 velocity–distance relation. The two lines use different corrections for the Sun's motion. (Note: The vertical units should be km s^{-1} .) (Figure from Hubble, *Realm of the Nebulae*, Yale University Press, New Haven, CT, ©1936.)

smaller telescopes, Hubble was so impressed with the results that Humason ended up as his assistant. Humason exposed and measured most of the photographic plates himself, and by 1934 the distances and velocities of 32 galaxies had been obtained. The expansion of the universe became an observational fact of life. Figure 27.8 shows the redshift of the H and K lines of Ca II for five galaxies.

Interestingly, the idea of a universal expansion was simultaneously being developed by theorists. In 1917, Willem de Sitter (1872–1935), a Dutch astronomer, used Einstein's theory of general relativity to describe a universe that was expanding.¹³ Although de Sitter's solution of Einstein's equations described an empty universe devoid of matter, it did predict a redshift that increased with the distance from the light's origin. Hubble was aware of de Sitter's work and, in his 1929 paper, stated that its “outstanding feature . . . is the possibility that the velocity–distance relation may represent the de Sitter effect.” Other theorists later found additional solutions that also indicated a universal expansion, but astronomers were unaware of their results until 1930. Einstein himself initially favored a static universe that was neither expanding nor contracting. However, the observations of Hubble and Humason forced Einstein to abandon this view in 1930. Figure 27.9 shows Einstein and Hubble at the Mt. Wilson Observatory.¹⁴

The Expansion of Space and the Hubble Flow

To understand what “the expansion of the universe” really means, suppose that Earth were to expand, doubling in size during an hour's time. Figure 27.10 shows the effect on the

¹³This and other models of the universe will be described in more detail in Chapter 29.

¹⁴During a tour of the observatory, Elsa Einstein was told how the telescope was used to explore the structure of the universe. She replied, “Well, well, my husband does that on the back of an old envelope.”

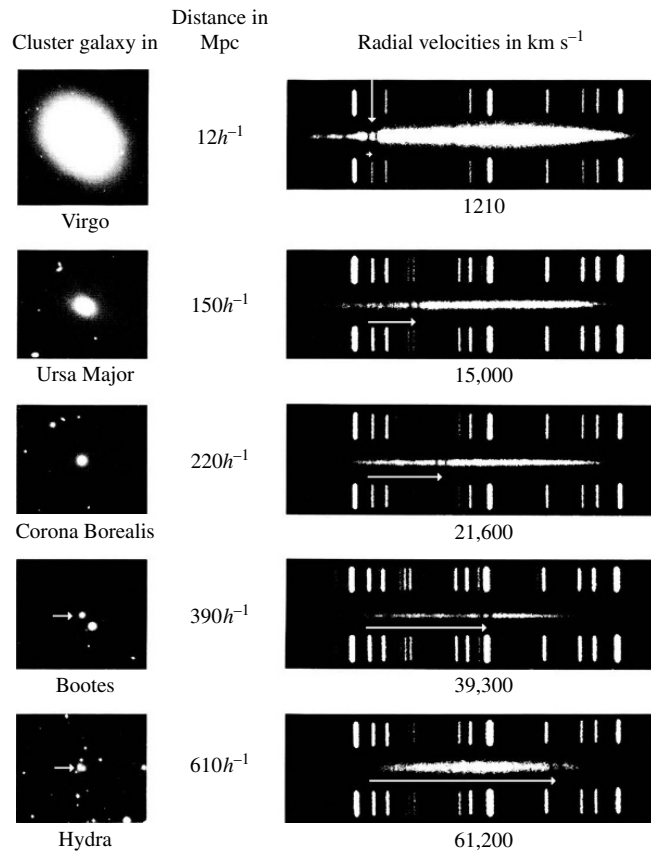


FIGURE 27.8 The appearance and redshifts of the H and K lines of calcium for five galaxies. (Courtesy of Palomar/Caltech.)

United States with two maps, aligned from the point of view of Salt Lake City. Initially, Yellowstone National Park is 500 km from Salt Lake City, Albuquerque is 1000 km away, and Minneapolis is at 2000 km. An hour later, Yellowstone is 1000 km away, Albuquerque is at 2000 km, and Minneapolis is at 4000 km, so a Salt Lake City resident would find that Yellowstone drifted away from her at 500 km hr^{-1} , Albuquerque moved away at 1000 km hr^{-1} , and Minneapolis receded at 2000 km hr^{-1} . Thus a recessional velocity that is proportional to distance is the natural result of an expansion that is both isotropic and homogeneous (same magnitude in every direction and at every location). Of course, observers in Yellowstone, Albuquerque, and Minneapolis would come to the same conclusion. Everyone involved in the expansion sees everyone else moving away with a velocity that obeys Hubble's law.

There is a vital distinction between the velocity of a galaxy through space (called its **peculiar velocity**) and its **recessional velocity** due to the expansion of the universe. As will be described in more detail in Section 29.1, a galaxy's recessional velocity is *not* due to its motion through space; instead, the galaxy is being carried along *with* the surrounding space



FIGURE 27.9 Einstein and Hubble at the Mt. Wilson Observatory. (Reproduced by permission of The Huntington Library, San Marino, CA.)

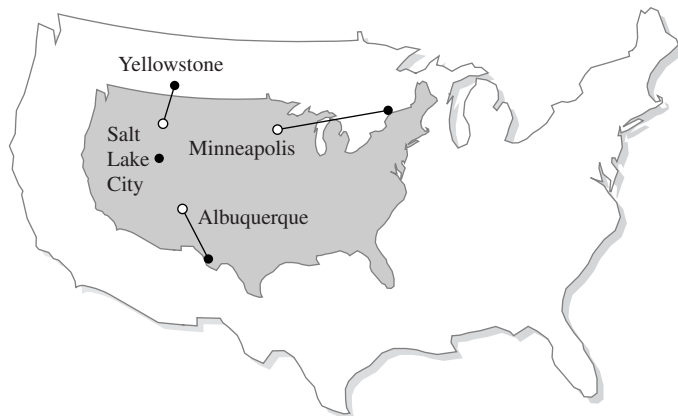


FIGURE 27.10 The effect of a doubling of the size of Earth, from the perspective of an astronomer in Salt Lake City.

as the universe expands. The motion of galaxies as they participate in the expansion of the universe is referred to as the **Hubble flow**. In the same manner, a galaxy's **cosmological redshift** is produced by the expansion as the wavelength of the light emitted by the galaxy is stretched along with the space through which the light travels.¹⁵ For this reason, the cosmological redshift is not related to the galaxy's recessional velocity by the Doppler shift equations derived in Section 4.3. Those equations were derived for a static, Euclidean spacetime and do not include the effects of the expanding, curved spacetime of our universe.

¹⁵The cosmological redshift will be derived in Section 29.4.

Nevertheless, astronomers frequently use Eq. (4.38) to translate a measured redshift, z , into the radial velocity a galaxy would have, *as if* it had a peculiar velocity (moving through space) instead of its actual recessional velocity (moving along with expanding space). For instance, the statement in Example 4.3.2 that “quasar SDSS 1030+0524 appears to be moving away from us at more than 96% of the speed of light” must be interpreted in this way. Furthermore, for $z \leq 2$, the distance estimate obtained by using Eq. (4.38) with Hubble’s law,

$$d \simeq \frac{c}{H_0} \frac{(z+1)^2 - 1}{(z+1)^2 + 1}, \quad (27.7)$$

differs from the actual proper distance by less than 5%.¹⁶ When $z \ll 1$, the expression for the distance assumes the nonrelativistic form

$$d = \frac{cz}{H_0}, \quad (27.8)$$

as would be found by using Eq. (4.39) with Hubble’s law. [As we will see in Section 29.4, Eq. (27.8) can lead to significant errors in the distance for roughly $z > 0.13$.] In Chapter 29, we will revisit the question of how distances are measured in an expanding universe.

It is important to realize that although the universe is expanding, this does not mean, for example, that the orbits of the planets around the Sun have been expanding. Gravitationally bound systems do not participate in the universal expansion. There is also no compelling evidence that the constants that govern the fundamental laws of physics (such as Newton’s gravitational constant, G) were once different from their present values. Thus the sizes of atoms, planetary systems, and galaxies have *not* changed because of the expansion of space (although the latter two may have certainly gone through evolutionary changes).

The Value of the Hubble Constant

In principle, Hubble’s law can be used to find the distance to any galaxy whose redshift can be measured. The stumbling block to carrying out this procedure has been the uncertainty in the Hubble constant. Through the end of the twentieth century, H_0 was known only to within a factor of two, between $50 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $100 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Historically, the difficulty in determining the value of H_0 arose from having to use remote galaxies for its calibration. A major source of the diverse values of H_0 obtained by different researchers lay in their choice and use of secondary distance indicators when measuring remote galaxies. There are also large-scale motions of galaxies relative to the Hubble flow that have yet to be sorted out, as will be described in Section 27.3.

In addition, there is a selection effect called a **Malmquist bias** that must be guarded against. This can occur when an astronomer uses a magnitude-limited sample of objects, looking only at those objects brighter than a certain *apparent* magnitude. At larger distances, only the intrinsically brightest objects will be included in the sample, which (if not properly corrected for) will skew the statistics.

¹⁶This statement assumes that the universe is “flat”; see Section 29.1 and Problem 29.58. Observations by missions such as Boomerang and WMAP indicate that space is very nearly flat.

As has been traditionally done in order to incorporate the uncertainty in H_0 , in this text we will define a dimensionless parameter, h , through the expression

$$H_0 = 100h \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (27.9)$$

Due to early uncertainty with H_0 , the value of h was known only to fall somewhere between about 0.5 and 1. However, as the end of the twentieth century approached, astronomers began to converge on the middle ground of values of H_0 .

The Big Bang

Since the universe is expanding, it must have been smaller in the past than it is now. Imagine viewing a filmed history of the universe, watching the galaxies fly farther and farther apart. Now run the film *backward*, in effect reversing the direction of time. Seen in reverse, all of the galaxies are approaching one another. According to the Hubble law, a galaxy twice as far away is approaching twice as fast. The inescapable conclusion is that all of the galaxies (*and all of space!*) will simultaneously converge to a single point. As everything in the universe rapidly converged (still running the film backward), it heated to extremely high temperatures. The expansion of the universe from a single point is known as the **Big Bang**.

The hot early universe was filled with blackbody radiation, and as the universe expanded that radiation cooled to become the **cosmic microwave background (CMB)**, microwaves that we detect arriving from all directions in the sky. Observations of the CMB provide convincing evidence for the hot Big Bang. The **Wilkinson Microwave Anisotropy Probe (WMAP)**,¹⁷ launched in 2001, studied the minute fluctuations (anisotropies) of the CMB. In 2003 the first WMAP results were released, the details of which will be discussed in Chapters 29 and 30. This ushered in a new era of precision cosmology, and the uncertainty in the Hubble constant (and h) fell from a factor of two to just 10%. The value of h determined by the WMAP data,¹⁸ and the standard value adopted in this text, is

$$[h]_{\text{WMAP}} = 0.71^{+0.04}_{-0.03}. \quad (27.10)$$

Throughout the remainder of this chapter, when a quantity involving h is evaluated, $[h]_{\text{WMAP}}$ will be assumed. In conventional units, the Hubble constant is

$$H_0 = 3.24 \times 10^{-18} h \text{ s}^{-1} \quad (27.11)$$

so

$$[H_0]_{\text{WMAP}} = 2.30 \times 10^{-18} \text{ s}^{-1}. \quad (27.12)$$

¹⁷David Wilkinson (1935–2002) made pioneering contributions to many CMB experiments, including COBE and WMAP. The MAP satellite was renamed “WMAP” after it was launched in 2001.

¹⁸Parameters published in Table 3 (“Best” Cosmological Parameters) of Bennett et al., 2003, such as H_0 , are placed inside a square bracket with a “WMAP” subscript, while quantities calculated using these parameters are not. A list of these parameters may be found in Appendix N. Although identified in the text as “WMAP parameters,” they are a combination of WMAP results with measurements from projects such as COBE, the Cosmic Background Imager (CBI), the Arcminute Cosmology Bolometer Array Receiver (ACBAR), and the 2dF Galaxy Redshift Survey (2dFGRS).

Thus a galaxy with a measured recessional velocity of 1000 km s^{-1} is at a distance of $d = v/H_0 = 10h^{-1} \text{ Mpc}$, or $d = 14.1 \text{ Mpc}$.

To estimate how long ago the Big Bang occurred, let t_H be the time that has elapsed since the Big Bang. This is the time required for a galaxy to travel to its present distance d while moving at its recessional velocity, v , as given by Hubble's law. Assuming (incorrectly) that v has remained constant,

$$d = v t_H = H_0 d t_H, \quad (27.13)$$

and so the **Hubble time** is

$$t_H \equiv \frac{1}{H_0} = 3.09 \times 10^{17} h^{-1} \text{ s} = 9.78 \times 10^9 h^{-1} \text{ yr}. \quad (27.14)$$

Using WMAP values,

$$t_H = 4.35 \times 10^{17} \text{ s} = 1.38 \times 10^{10} \text{ yr}. \quad (27.15)$$

Thus, as a rough estimate, the age of the universe is about 13.8 Gyr.

27.3 ■ CLUSTERS OF GALAXIES

Astronomers hold as a fundamental tenet that, on the largest scales, the universe is both homogeneous and isotropic, appearing the same at all locations and in all directions (an assumption called the **cosmological principle**). However, this is certainly not the case at smaller scales.

The Classification of Clusters

As has been mentioned a number of times already in this text, galaxies are not strewn randomly throughout the universe; instead, nearly all galaxies are found in associations, either in **groups** or in **clusters**.¹⁹ In both types, the galaxies are gravitationally bound to one another and orbit the system's barycenter (center of mass).

Groups generally have less than 50 members and are about $1.4h^{-1} \text{ Mpc}$ across. The galaxies of a group have a velocity dispersion of about 150 km s^{-1} , and the mass of an average group is on the order of $2 \times 10^{13} h^{-1} M_\odot$, obtained from the virial theorem (Eq. 25.13). Furthermore, the mass-to-light ratio of a typical group is about $260h M_\odot/L_\odot$, which is indicative of large amounts of dark matter.

Clusters, on the other hand, may contain from approximately 50 galaxies (a **poor** cluster) to thousands of galaxies (a **rich** cluster), within a region of space about $6h^{-1} \text{ Mpc}$ in diameter. The individual galaxies in a cluster move more rapidly with respect to other members than do galaxies residing in a group. The characteristic velocity dispersion of a cluster is 800 km s^{-1} , possibly exceeding 1000 km s^{-1} for very rich clusters. A typical cluster's virial mass is around $1 \times 10^{15} h^{-1} M_\odot$, and its mass-to-light ratio is roughly $400h M_\odot/L_\odot$, again

¹⁹Some galaxies are not affiliated with a known group or cluster, but they represent rare exceptions.

indicative of large amounts of dark matter. Clusters of galaxies are further classified as **regular** (spherical and centrally condensed) and **irregular**.

In this section, we will describe how groups of galaxies, clusters of galaxies, and clusters of clusters (called **superclusters**) make up the basic structure of the universe.

The Local Group

As shown in Fig. 27.11, about 35 galaxies are known to lie within roughly 1 Mpc of the Milky Way. This collection of galaxies is known as the **Local Group**. Its **zero velocity surface** (where the galaxies would eventually turn around if their motions were suddenly directed outward) is about 1.2 Mpc from the barycenter. Its most prominent members are its three spiral galaxies: our Galaxy, M31 (the Andromeda galaxy), and M33 (in the constellation Triangulum). The Large and Small Magellanic Clouds are the next most luminous; they are two of the 13 irregular galaxies in the Local Group. [The LMC and SMC are the two prominent features below the Galactic equator in Fig. 6.28(c).] The remaining galaxies are dwarf ellipticals or dwarf spheroidals, which are very small and quite faint. It is obvious from Fig. 27.11 that many of these galaxies have accumulated around the Milky Way and Andromeda galaxies, which are on opposite sides of the Local Group, about $r = 770$ kpc apart.²⁰ (Even at this distance, M31 is about 2.5° wide on the sky, five times the Moon's diameter.) In addition to the Magellanic Clouds, nine of the dwarf ellipticals and dwarf

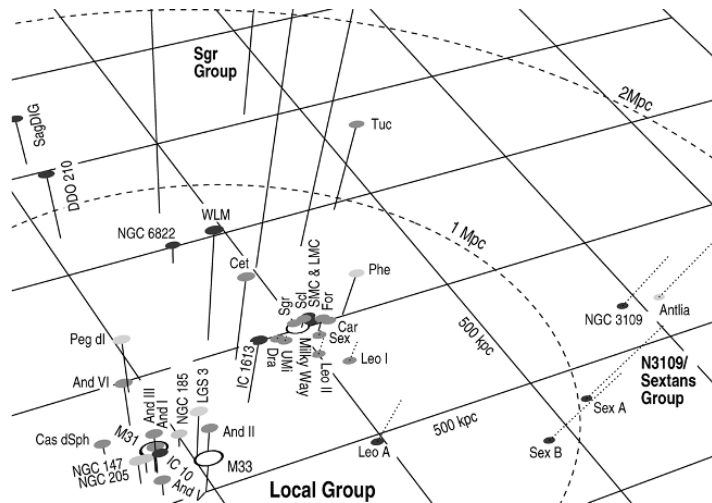


FIGURE 27.11 The Local Group of galaxies. The dashed circles are centered on the barycenter of the Local Group and have radii of 1 and 2 Mpc. (The barycenter is 462 kpc from us in the direction of galactic coordinate $\ell = 121.7^\circ$ and $b = -23.1^\circ$.) Solid lines indicate distances above the (arbitrarily chosen) plane, and dashed lines show distances below the plane. (Figure from Grebel, “The Local Group,” in *Microlensing 2000: A New Era of Microlensing Astrophysics*, A. S. P. Conference Series, 239, 280, 2001.)

²⁰This accumulation can also be seen in Hubble’s own velocity–distance diagram, Fig. 27.7; the Local Group is represented by the eight data points near the origin.

spheroidals are situated near the Milky Way. Interestingly, several of them are found in the Magellanic Stream, the long ribbon of neutral hydrogen gas that was tidally stripped from the Magellanic Clouds some 200 million years ago. It should also be noted that all three spiral galaxies have warped disks that resemble an integral sign ($\oint$) when viewed edge on (recall the map of the H I warp in our Galaxy shown in Fig. 24.10). In fact, the line of sight from Earth passes twice through some parts of the disk of M33.

The Andromeda and Milky Way galaxies are approaching each other with a relative velocity of $v = 119 \text{ km s}^{-1}$.²¹ Apparently, the gravitational attraction between them has overcome their tendency to expand along with the Hubble flow. As a result, they will collide in approximately $t_c = r/v = 6.3$ billion years.²² Indeed, astronomers have found that the entire Local Group is still in a state of collapse.

Because the Milky Way and Andromeda dominate the Local Group, producing some 90% of its luminosity, their motion toward each other provides an opportunity to estimate their combined mass. We can assume that following the Big Bang, the two newly formed galaxies were initially moving apart.²³ At some time in the past, their gravitational attraction halted and reversed their original recessional motion. The Milky Way and M31 are therefore in orbit about one another with an eccentricity $e \simeq 1$ (i.e., on a collision course). From conservation of energy, the orbital speed and separation are related by

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right),$$

(Eq. 2.36), where M is the total mass of the two galaxies. The orbit's semimajor axis, a , is related to its period, P , by Kepler's third law, Eq. (2.37):

$$P^2 = \frac{4\pi^2}{GM} a^3.$$

Combining these relations to eliminate a leads to

$$v^2 - \frac{2GM}{r} + \left(\frac{2\pi GM}{P} \right)^{2/3} = 0. \quad (27.16)$$

In this equation,²⁴ $r = 770 \text{ kpc}$ and $v = 119 \text{ km s}^{-1}$. To obtain the total mass M , we need an estimate of the orbital period, P . When the galaxies do meet in the future, they will have returned to their initial configuration, and one orbital period will have elapsed since the Big Bang. Let us therefore use

$$P = t_H + t_c,$$

²¹M31 is approaching the Sun with a speed of 297 km s^{-1} , but a large fraction of this is due to the Sun's orbital motion around the Milky Way.

²²This suggests the possibility that they might then merge to form a giant elliptical galaxy.

²³Remember, in the absence of peculiar velocities, an *expansion* means that every point in space moves directly away from every other point.

²⁴Equation (27.16) is a cubic equation for $M^{1/3}$ that has one real root and two complex roots. The real root is the one with physical significance.

where $t_c = r/v$ is roughly the time until they collide. This is an overestimate of the period, because the Milky Way and M31 will accelerate as they approach their rendezvous. For the WMAP value $h = 0.71$ (so $H_0 = 71 \text{ km s}^{-1} \text{ Mpc}^{-1}$), the total mass is found to be $M = 7.9 \times 10^{42} \text{ kg} = 4.0 \times 10^{12} M_\odot$.

This estimate of the mass of the Local Group is much more than the luminous mass observed in these galaxies. The B -band luminosity of the Milky Way is about $2.3 \times 10^{10} L_\odot$ (see Table 24.1), and M31 is roughly twice as bright. Thus, the mass-to-light ratio is estimated to be $M/L = 57 M_\odot/L_\odot$. Note that a smaller period gives a larger mass. This means that, because the value of P was overestimated, M/L has been *underestimated*. These numbers are much larger than the value of $M/L \simeq 3 M_\odot/L_\odot$ for the luminous matter in the Milky Way's thin disk and central bulge (Section 24.2). Our WMAP result of $M/L \simeq 57 M_\odot/L_\odot$ is consistent with the estimates quoted in Section 24.2 when the Milky Way's dark halo is included. Apparently, astronomers have seen less than approximately 10% of the matter that makes up the Milky Way and Andromeda galaxies.

A careful study of the Magellanic Stream supports this conclusion. The tip of this flow of neutral hydrogen gas is plunging toward the Galaxy with a speed of 220 km s^{-1} . Researchers attempting to reproduce the dynamics of the stream obtain the best results with a model that has a halo of dark matter extending at least 100 kpc from the Galaxy. The Galactic mass-to-light ratio for these models exceeds $80 M_\odot/L_\odot$.

Other Groups within 10 Mpc of the Local Group

The Local Group is indeed a spatial concentration of galaxies. Although there are some 35 galaxies in the Local Group within a volume of space about a megaparsec across, the next nearest galaxies are found in the Sculptor group (six members, 1.8 Mpc away) and in the M81 group (eight members, 3.1 Mpc away). The six spiral galaxies of the Sculptor group span about 20° on the night sky. You could just cover them with your outstretched hand held at arm's length.²⁵ Other associations within 10 Mpc include the Centaurus group (17 members, 3.5 Mpc away) and the M101 group (5 members, 7.7 Mpc away). The M66 and M96 groups are separated by only 7° in the sky. Together they contain about 10 galaxies, and both are about 9.4 Mpc from Earth. And finally, the last group within 10 Mpc of Earth is the NGC 1023 group (6 members, 9.5 Mpc away). Altogether, there are about twenty small groups of galaxies that are closer to us than the Virgo cluster. Most galaxies are found in such small groups and poor clusters. It is estimated that, at most, 20% of all galaxies inhabit rich clusters like the Virgo cluster.

Figure 27.12 shows the positions of the nearest galaxies. Centered on the Milky Way, the diagram reaches out to just past the M101 group. There are many more dwarfs than there are giant galaxies, and both types are found generally in the same regions. There are also some extended volumes of space, termed **voids**, where galaxies are apparently absent. Another striking feature is that the galaxies show a tendency to lie along a two-dimensional plane ($SGZ = 0$) that forms part of the Local Supercluster of galaxies centered on the Virgo cluster (to be discussed later).²⁶

²⁵It is fascinating to realize that many astronomical objects occupy a significant area of the sky. If only our eyes had the sensitivity to see them!

²⁶“SG” denotes a supergalactic coordinate system created by de Vaucouleurs.

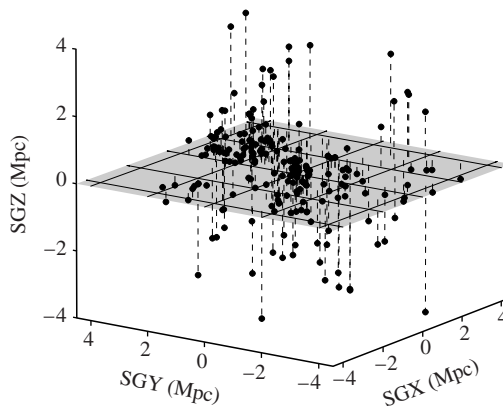


FIGURE 27.12 Galaxies near the Milky Way. Most of the positions (except for M31 and galaxies in its subgroup) were found using redshifts and the Hubble law. The *SGY*-axis points toward the Virgo cluster.

The Virgo Cluster: A Rich, Irregular Cluster

The **Virgo cluster** of galaxies was first recognized in the eighteenth century by William Herschel. Located where the constellations of Virgo and Coma Berenices (Berenice's Hair) meet, the Virgo cluster covers a $10^\circ \times 10^\circ$ region of the sky.²⁷ The center of the cluster is located about 16 Mpc from Earth.²⁸ This rich, irregular cluster is a sprawling collection of approximately 250 large galaxies and more than 2000 smaller ones, contained within a region about 3 Mpc across. Interestingly, at least seven of these galaxies display blueshifted spectral lines, as their peculiar velocities of approach overpower the receding Hubble flow.

Like most irregular clusters, the Virgo cluster is made up of all types of galaxies. Although the four brightest are giant elliptical galaxies, ellipticals make up only 19% of the cluster's 205 brightest galaxies. Spirals dominate overall, accounting for 68% of the top 205. However, roughly equal numbers of spirals and dwarf ellipticals are found within 6° of the cluster's center, with the ellipticals becoming increasingly common near the center. In fact, the center of the Virgo cluster is dominated by three of the cluster's four giant elliptical galaxies (M84, M86, and M87; see Figs. 27.13 and 28.10). The diameters of these galaxies are comparable to the distance between our Galaxy and Andromeda, so each of these giant ellipticals is nearly the size of the entire Local Group.

M87, a giant E1 elliptical, is the largest and brightest galaxy in the Virgo cluster. Like many other luminous elliptical galaxies, it contains roughly $10^{10} M_\odot$ of hot ($\sim 10^7$ K) gas that has accumulated through normal stellar mass loss mechanisms. The gas loses energy by the free-free emission of X-ray photons. This thermal bremsstrahlung process, first discussed in Section 9.2, produces a characteristic spectrum that can be readily identified

²⁷Berenice, wife of the Egyptian king Ptolemy III, offered her golden tresses as a sacrifice to the goddess Aphrodite in gratitude for her husband's victory over the Assyrians.

²⁸This value of the distance to the Virgo cluster is in the best agreement with the values in Table 27.1; see Problem 27.19.



FIGURE 27.13 The center of the Virgo cluster, showing the giant ellipticals M84 (right) and M86 (center). (Courtesy of National Optical Astronomy Observatories.)

and used to estimate the mass of the galaxy [see Eq. (27.18) and Fig. 27.17]. Although the analysis is complicated, observations of the spectrum of the gas can provide an idea of how its temperature, $T(r)$, and mass density, $\rho(r)$, vary with r , the distance from the center of the galaxy. Fortunately, the nearly spherical shape of M87 simplifies the geometry. The gas is in hydrostatic equilibrium (to a good approximation), so from Eq. (10.6),

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2}.$$

Substituting the ideal gas law, Eq. (10.11), for the pressure, P , assuming constant μ , and solving for the interior mass, M_r , produce

$$M_r = -\frac{kTr}{\mu m_H G} \left(\frac{\partial \ln \rho}{\partial \ln r} + \frac{\partial \ln T}{\partial \ln r} \right). \quad (27.17)$$

All of the quantities on the right-hand side can be roughly evaluated from observations of the X-rays emitted by the hot gas, and used to calculate the interior mass on the left-hand side. Note that M_r is the *total* interior mass, both luminous and dark. (The derivatives are themselves negative, so M_r is positive.)

One result for M87 shows that M_r increases *linearly* with radius out to about 300 kpc. This is the same signature seen for the distribution of dark matter in spiral galaxies. The total mass contained within $r = 300$ kpc is $M_r \simeq 3 \times 10^{13} M_\odot$, with a central density of $1.5 \times 10^{-2} M_\odot \text{ pc}^{-3}$. The corresponding mass-to-light ratio is $M/L \simeq 750 M_\odot/L_\odot$. This is 250 times the mass-to-light ratio for the stars in the Milky Way's Galactic bulge and thin disk (Section 24.2) and indicates that over 99% of M87's mass is dark matter. As we will see in Section 28.1, M87 is *not* a typical elliptical galaxy. However, studies of other elliptical

galaxies have found that many of them also contain large amounts (90% or more) of dark matter.

The Coma Cluster: A Rich, Regular Cluster

The nearest rich, regular cluster of galaxies is the **Coma cluster**. It is about 15° north (in declination) of the Virgo cluster, in the constellation Coma Berenices. Recall from Example 27.1.4 that the Coma cluster is about 5.4 times farther away than is the Virgo cluster, or about 90 Mpc from Earth. The cluster's angular diameter is about 4° , which corresponds to a linear diameter of 6 Mpc. The Coma cluster consists of perhaps 10,000 galaxies, most of them dwarf ellipticals that are too faint to be seen. Generally, in a rich, regular cluster the vast majority of the galaxies are ellipticals and S0's. This is the case for the Coma cluster. It contains over 1000 bright galaxies, but only 15% of them are spirals and irregulars. At the cluster's center are two large, luminous cD ellipticals; see Fig. 26.1.

Evidence for the Evolution of Galaxies

Recall that the predominance of early-type galaxies in a cluster may be due to the increased likelihood of interactions (the morphology–density relation). Perhaps in the past, more spirals existed in the Coma cluster and other rich, regular clusters, but tidal interactions and mergers destroyed the spiral morphology, leaving ellipticals and S0's.²⁹ Alternatively, early-type galaxies may simply form as a result of hierarchical mergers of protogalaxies near the bottom of the cluster's gravitational well, in regions where dark matter would naturally tend to collect.

In support of the interaction view, Fig. 27.14 shows the center of the very rich cluster CL 0939+4713, with a redshift of $z = 0.41$, implying a distance of $1230h^{-1} = 1730$ Mpc. We are seeing this cluster as it was about $4.0h^{-1} = 5.6$ billion years ago, when many late-type spirals were found in this central region. This is unlike the center of any high-density cluster belonging to the present epoch. Most of the spirals in Fig. 27.14 show at least some evidence of an interaction. The frequency of interactions and mergers is perhaps an order of magnitude greater than would be found in a contemporary rich cluster.

Looking even more deeply into the universe, astronomers have discovered a remote cluster in the constellation Serpens at $z \approx 1.2$, or $1970h^{-1} = 2770$ Mpc. This cluster, shown in Fig. 27.15, is seen as it was about $6.4h^{-1} = 9.0$ billion years ago. It contains apparently mature ellipticals but very few normal spirals. Instead, there is an abundance of odd galactic fragments having a bluish color (a signature of active star formation; recall the Butcher–Oemler effect). It is not clear whether these are the debris of former spirals, or spirals still in the process of coalescing. Further observations should reveal more details about the formation and disruption of spiral galaxies in the dynamic environment of such crowded young clusters.

A Preponderance of Matter between the Galaxies

In 1933 Fritz Zwicky measured the radial velocities of galaxies in the Coma cluster from their Doppler-shifted spectra. From those observations he calculated the dispersion in their

²⁹The greater frequency of spirals in young clusters was discussed at the end of Section 26.2.

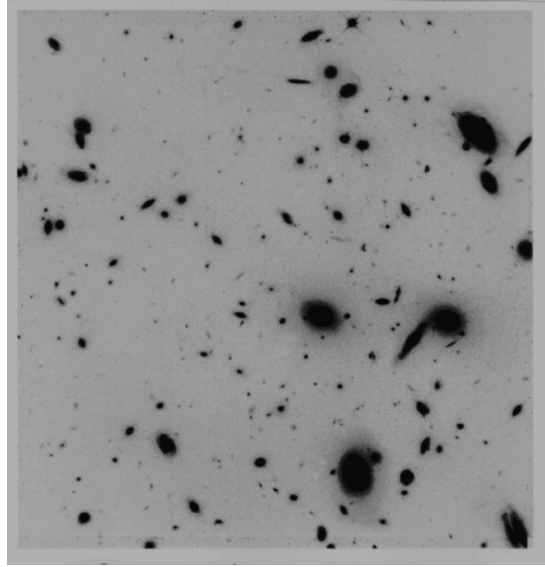


FIGURE 27.14 An HST image of the center of the rich cluster CL 0939+4713. (Figure from Dressler et al., *Ap. J. Lett.*, 435, L23, 1994.)

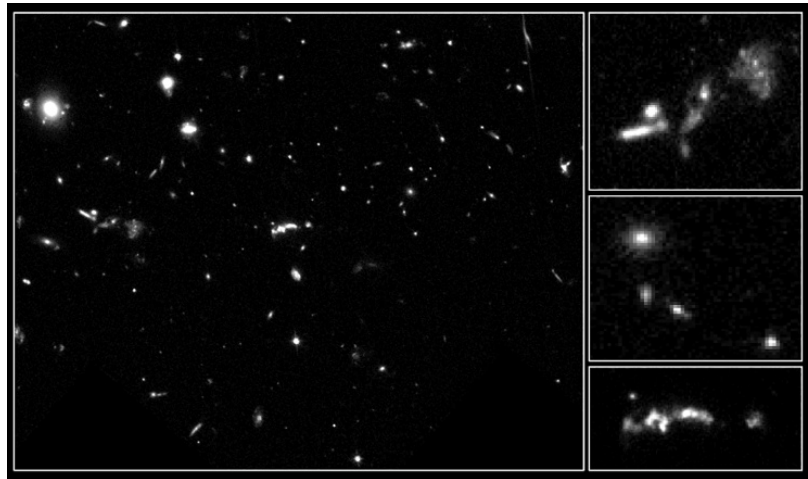


FIGURE 27.15 An HST view of a young cluster of galaxies, centered on the peculiar radio galaxy 3C 324 (also seen at bottom right) at $z = 1.2$. At center right is a pair of normal-appearing elliptical galaxies with a few faint companions, while at top right are some galactic fragments that may become, or may have once been, spirals. [Courtesy of Mark Dickinson (STScI) and NASA.]

radial velocities, now known to be $\sigma_r = 977 \text{ km s}^{-1}$. Zwicky then used the virial theorem to estimate the mass of the Coma cluster. Overall, the Coma cluster's intensity profile, $I(r)$, follows a characteristic $r^{1/4}$ law (Eq. 24.13) like those that describe the intensity profiles of the Milky Way's bulge and halo, and elliptical galaxies. Presumably, like the stars that make up elliptical galaxies and the spheroidal components of spirals, the galaxies in the Coma cluster have become dynamically relaxed and are in an equilibrium configuration. This makes the virial theorem an appropriate method to use.

Example 27.3.1. In the case of the Coma cluster, the dispersion in the radial velocity is 977 km s^{-1} . With the cluster radius $R = 3 \text{ Mpc}$, Eq. (25.13) leads to a mass of

$$M \approx \frac{5\sigma_r^2 R}{G} = 3.3 \times 10^{15} M_\odot$$

for the Coma cluster. Since the visual luminosity of the Coma cluster is about $5 \times 10^{12} L_\odot$, the mass-to-light ratio of the cluster is $M/L \approx 660 M_\odot/L_\odot$.³⁰

Zwicky understood the significance of this result and stated that in the Coma cluster, “the total mass . . . considerably exceeds the sum of the masses of individual galaxies.” He realized that there is not enough visible mass to bind the cluster together. If not for the presence of a large amount of unseen matter, the galaxies in the Coma cluster would have dispersed long ago. Forty years later, when the flat rotation curve of the Andromeda galaxy was measured, other astronomers also realized the significance of Zwicky's result.

The Hot, Intracluster Gas

A portion of Zwicky's “missing mass” was discovered with the High Energy Astronomical Observatory (HEAO) series of satellites that were first launched in 1977. They revealed that many clusters of galaxies emit X-rays from much of the cluster's volume. These satellites, together with optical observations, indicated that clusters of galaxies contain an **intracluster medium**. The intracluster medium has two components. One is a diffuse, irregular distribution of stars. The other component is a hot **intracluster gas** that is distributed more or less homogeneously, occupying the space between the galaxies and filling the cluster's gravitational potential well. The X-ray luminosities lie in the range 10^{36} to 10^{38} W , with richer clusters shining more brightly in X-rays. The core of the Virgo cluster contains about $5 \times 10^{13} M_\odot$ of X-ray emitting gas, and the core of the Coma cluster holds $3 \times 10^{13} M_\odot$. Figure 27.16 shows the X-ray emission from the hot gas that fills the Coma cluster. Typically the mass of the gas is several times greater than the combined mass of all stars in the cluster's galaxies.

Thermal bremsstrahlung, the same mechanism that produces the X-rays coming from the hot gas within individual galaxies (described earlier for M87 in the Virgo cluster), is at work here as well. For fully ionized hydrogen gas, the energy emitted per unit volume per unit time between frequencies ν and $\nu + d\nu$ is given by

$$\ell_\nu d\nu = 5.44 \times 10^{-52} \left(4\pi n_e^2 \right) T^{-1/2} e^{-h\nu/kT} d\nu \text{ W m}^{-3}, \quad (27.18)$$

³⁰A more detailed calculation results in a mass-to-light ratio about half as large as our estimate.

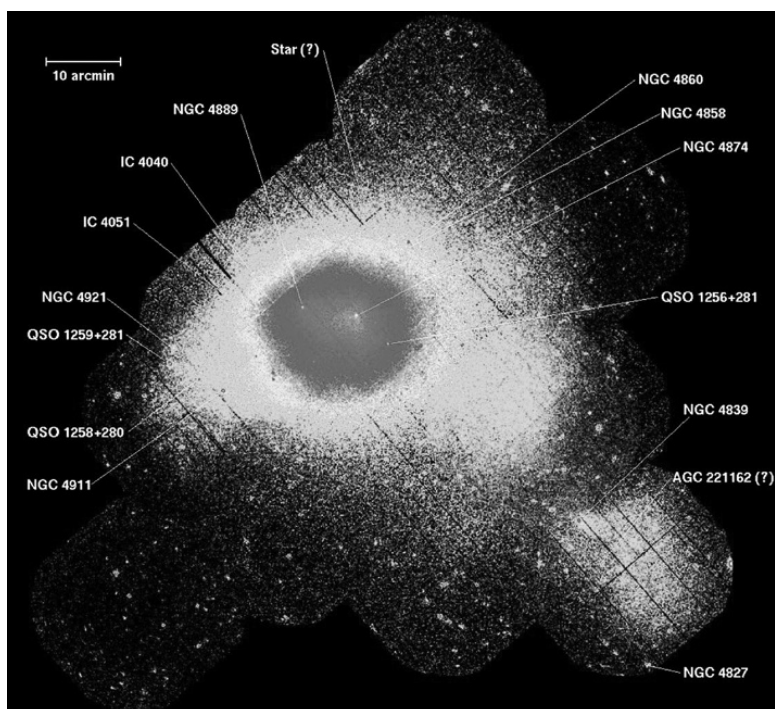


FIGURE 27.16 An X-ray Multi-Mirror (XMM) telescope image of the Coma cluster taken by the European Space Agency's XMM-Newton space observatory. The mosaic image shows X-rays in the 0.3–2.0-keV range from the Coma cluster's intracluster gas and occupies a field of view of $1.5^\circ \times 1.33^\circ$. The intensity of the X-rays increases continuously toward the center of the cluster. The part of the Coma cluster shown in Fig. 26.1 occupies the most intense central area of this image. The irregular shape of the emission indicates that the Coma cluster may have formed from the merger of several subclusters. (Image courtesy of U. Briel, Max Planck Institut für extraterrestrische Physik, Garching, Germany, and the European Space Agency.)

where T is the gas temperature and n_e is the number density of free electrons. One version of this spectrum for the Coma cluster is shown in Fig. 27.17. The total amount of energy emitted per second per unit volume at all frequencies (the *luminosity density*, $\mathcal{L}_{\text{vol}}$) is obtained by integrating ℓ_ν over frequency. This results in

$$\mathcal{L}_{\text{vol}} = 1.42 \times 10^{-40} n_e^2 T^{1/2} \text{ W m}^{-3}. \quad (27.19)$$

Example 27.3.2. Equation (27.19) can be used to estimate the mass of the Coma cluster's intracluster gas. For simplicity, we will assume that the cluster can be modeled as an isothermal sphere of hot ionized hydrogen gas. Figure 27.16 indicates that the central core radiates most strongly, so we will take the radius to be $R = 1.5$ Mpc (one-half of the cluster's actual radius). The gas is optically thin, meaning that any photon emitted along the line of sight can be observed. The gas temperature comes from the X-ray spectrum of the gas,

continued

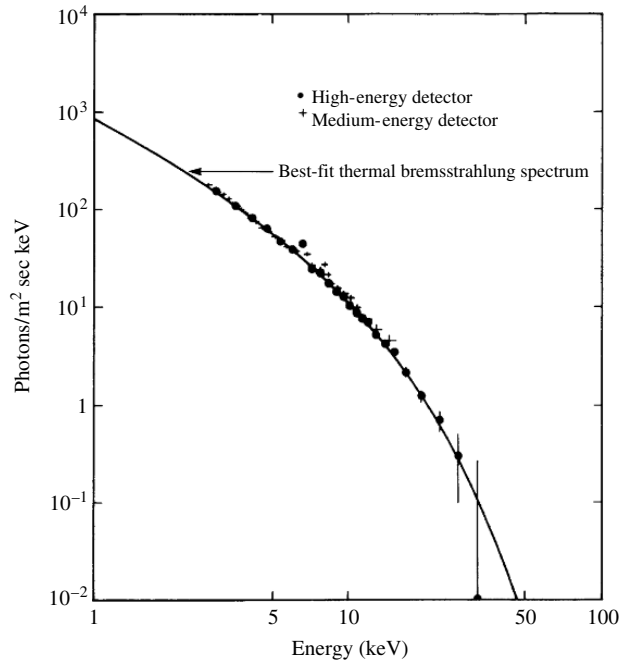


FIGURE 27.17 Thermal bremsstrahlung spectrum (line) for 88 million K. The points are observations of X-rays from the Coma cluster's intracluster gas. Photon energy is plotted on the horizontal axis. (Figure adapted from Henriksen and Mushotzky, *Ap. J.*, 302, 287, 1986.)

shown in Fig. 27.17. Using Eq. (27.18), the best fit to the data (the dots on the figure) is obtained with a temperature of 8.8×10^7 K.

Using Eq. (27.19), L_x can be written as

$$L_x = \frac{4}{3} \pi R^3 \mathcal{L}_{\text{vol}}. \quad (27.20)$$

Since the X-ray luminosity of the gas is $L_x = 5 \times 10^{37}$ W, the value of n_e , the number of free electrons per m^{-3} , is

$$n_e = \left[\frac{3L_x}{4\pi R^3 T^{1/2} (1.42 \times 10^{-40} \text{ W m}^{-3})} \right]^{1/2} = 300 \text{ m}^{-3}.$$

The intracluster gas is several million times less dense than the giant molecular clouds described in Section 12.1, for which $n_{\text{H}_2} \sim 10^8$ to 10^9 m^{-3} .

For ionized hydrogen, there is one proton for every electron, so the total mass of the gas is

$$M_{\text{gas}} = \frac{4}{3} \pi R^3 n_e m_H = 1.05 \times 10^{14} M_{\odot},$$

a slight overestimate of the value of $3 \times 10^{13} M_{\odot}$ quoted previously.

Let's compare the foregoing mass of the intracluster gas with the luminous mass of the Coma cluster. Using the mass-to-light ratio for the stars in the Milky Way's Galactic bulge and thin disk, $M/L \simeq 3 M_{\odot}/L_{\odot}$, and the visual luminosity of the Coma cluster, $L_V = 5 \times 10^{12} L_{\odot}$, the visible mass of the Coma cluster is approximately $1.5 \times 10^{13} M_{\odot}$. According to this estimate, there is roughly 7 times more intracluster gas in the Coma cluster than there is in its galaxies' stars. However, the $10^{14} M_{\odot}$ of intracluster gas is still only a few percent of the total mass of the Coma cluster.

The X-ray spectrum of the intracluster gas also displays emission lines of highly ionized iron (e.g., Fe XXV and Fe XXVI), silicon, and neon. This indicates that the gas has been processed through the stars in the cluster's galaxies, becoming enriched in heavy elements via stellar nucleosynthesis. How, then, did so much mass escape from the cluster's galaxies?

Mergers must have occurred more frequently in the early history of the cluster. When the cluster was still forming during its first few billion years, it was more dispersed. According to the virial theorem, expressed in the form of Eq. (25.13), the cluster's galaxies would have been moving more slowly, thereby increasing the dynamical friction and the chance of a merger. The large amounts of intracluster gas found in many rich clusters were probably ejected during these early galactic interactions, or by bursts of star formation. Once this process was under way, it would have been enhanced by ram-pressure stripping. When a galaxy moves at several thousand kilometers per second through the intracluster gas, it encounters a furious wind that is capable of stripping its gas away.

The Existence of Superclusters

Next in the hierarchy of galactic clustering is the **supercluster**. As the name suggests, this is a clustering of clusters on a grand scale (up to about 100 Mpc). It may be that virtually every galaxy belongs to a supercluster; certainly every known rich cluster is found in one.

The Virgo cluster is near the center of the **Local Supercluster**. This disk of clusters has the shape of a flattened ellipsoid, a pancake that contains most of the galaxies lying within about 20 Mpc of the Virgo cluster (including the Local Group, which is located near the edge of the supercluster). Figure 27.18 shows the Local Supercluster as the linear grouping of galaxies extending from the center of the figure. The view that is depicted is parallel to the plane of the supercluster, so you are looking at the pancake edge-on. Compare this view with Fig. 27.12 and note that the concentration of galaxies along a plane continues in Fig. 27.18. Of the other known superclusters, two of the most prominent are the Perseus–Pisces supercluster, named for the northern constellations in which it is found, and the Hydra–Centaurus supercluster in the southern sky. The Perseus–Pisces supercluster is about $50h^{-1}$ Mpc away and has a threadlike appearance, a planar, filamentary structure that is about $40h^{-1}$ Mpc long; see Fig. 27.25. The Hydra–Centaurus supercluster is at $30h^{-1}$ Mpc, in roughly the opposite direction.

Because the Local Group resides near the edge of the Local Supercluster, it might be expected that the supercluster's gravitational pull would be detectable. In 1958 de Vaucouleurs found that the Local Group is moving away less rapidly from the Virgo cluster than would be expected solely from the expansion of the universe. At a distance of 16 Mpc, a pure Hubble flow would produce a recessional velocity of $v = H_0 d = 1600h \text{ km s}^{-1}$. The difference between the actual velocity of the Local Group relative to the Virgo cluster and

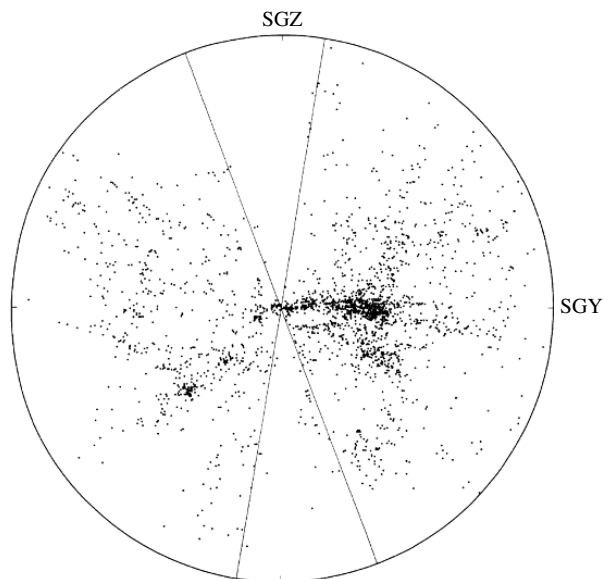


FIGURE 27.18 The distribution of 2175 bright galaxies out to roughly 50 Mpc. The Local Supercluster extends to the right, with the Milky Way (at center) located near the edge of the supercluster. The plane of the Milky Way bisects the “empty” slices; galaxies within these slices are hidden from view by Galactic dust and gas (the zone of avoidance). (Figure from Tully, *Ap. J.*, 257, 389, 1982.)

the Hubble flow, called the **Virgocentric peculiar velocity** of the Local Group, is estimated to be $168 \pm 50 \text{ km s}^{-1}$ by Sandage and Tammann (1990). An argument similar in spirit to that used for the approaching paths of the Milky Way and M31 can now be invoked to estimate the mass and the mass-to-light ratio of the Local Supercluster. The results show that the supercluster’s mass is about $8 \times 10^{14} h^{-1} M_{\odot}$. The corresponding mass-to-light ratio is $M/L \simeq 400h M_{\odot}/L_{\odot}$, providing more evidence for a preponderance of dark matter.

Large-Scale Motions Relative to the Hubble Flow

The Virgocentric peculiar velocity is a minor perturbation in a much larger scale inhomogeneity in the Hubble flow. As we will see in Section 29.2, there is a large-scale streaming motion (relative to the Hubble flow) that carries the Milky Way, the Local Group, the Virgo cluster, and thousands of other galaxies through space in the direction of the constellation Centaurus. The peculiar velocity of the Local Group relative to the Hubble flow is 627 km s^{-1} . This riverlike motion of thousands of galaxies extends at least $40h^{-1} \text{ Mpc}$ both upstream and downstream. Astronomers would like to use this streaming to deduce the location(s) of the mass, visible or dark, capable of exerting such an immense gravitational tug. The Hydra–Centaurus supercluster, which is in the direction of the flow, is *not* responsible, since it too is moving along with the flow of the rest of the galaxies. This implies that the source of the motion lies beyond that supercluster.

In the 1980s, American astronomers Alan Dressler and Sandra Faber calculated the presence of a **Great Attractor** (GA), a diffuse collection of clusters spread over a wide

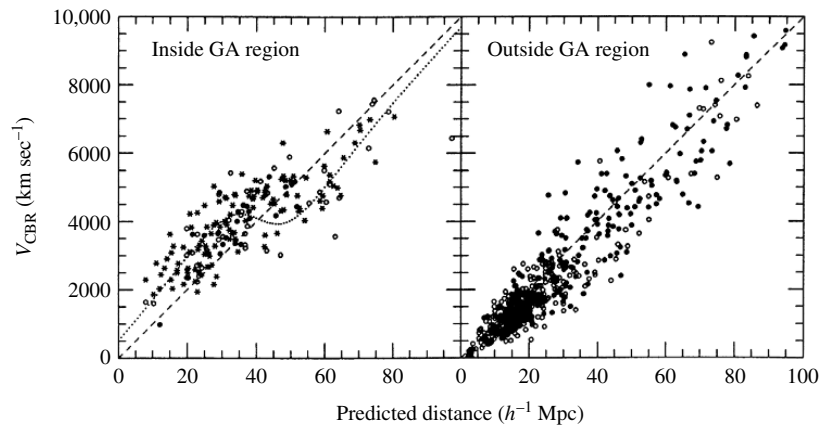


FIGURE 27.19 Left: velocities of galaxies in the Centaurus region, compared with the Hubble flow (dashed line). The dotted line shows the theoretical variation in velocity produced by a model of the Great Attractor. The Hydra–Centaurus supercluster is centered at about $30h^{-1}$ Mpc. Right: comparison figure for galaxies observed in another direction. (Figure adapted from Dressler and Faber, *Ap. J. Lett.*, 354, L45, 1990.)

(60°) region of the sky. According to their calculations, the GA lies in the same plane as the Local Supercluster, about $42h^{-1}$ Mpc away in the direction of $\ell = 309^\circ$, $b = 18^\circ$ (Galactic coordinates) in the constellation Centaurus. The mass of the Great Attractor is estimated to be about $2 \times 10^{16}h^{-1} M_\odot$, but there are just 7,500 galaxies known in that region—too few to account for so much mass. This suggests that approximately 90% of the mass of the GA may be in the form of dark matter. Alternatively, because the Great Attractor lies behind the plane of our galaxy and is obscured by dust, there could be a large hidden supercluster of galaxies in that direction. If so, it would be centered on a cluster of galaxies called Abell 3627.

Figure 27.19 shows the velocities of E and S0 galaxies in the region of the Great Attractor; the figure's dotted line is the galactic velocity field calculated for a specific model of the GA. On the near side of the GA, the velocities clearly exceed that of the unperturbed Hubble flow (the diagonal dashed line in the figure). There are hints of a “back flow” of the farside of the GA (velocities less than the Hubble flow), but this has been disputed by other investigators.³¹ These astronomers believe that the velocity excess actually continues *beyond* the proposed location of the GA and suggest other (or additional) concentrations of matter. Whether or not the Great Attractor exists as described, the existence of large-scale streaming motions relative to the Hubble flow seemed undeniable.

Another possibility is that the Great Attractor is not solely responsible for the large-scale streaming motion. The **Shapley concentration** of galaxies is probably the most massive collection of galaxies in our neighborhood of the universe; see Fig. 27.20. With a core consisting of a gravitationally bound concentration of some 20 rich clusters of galaxies, the Shapley concentration has a mass of a few $\times 10^{16}h^{-1} M_\odot$ and lies within 10° of the

³¹See, for example, a similar diagram in Mathewson, Ford, and Buchhorn (1992).

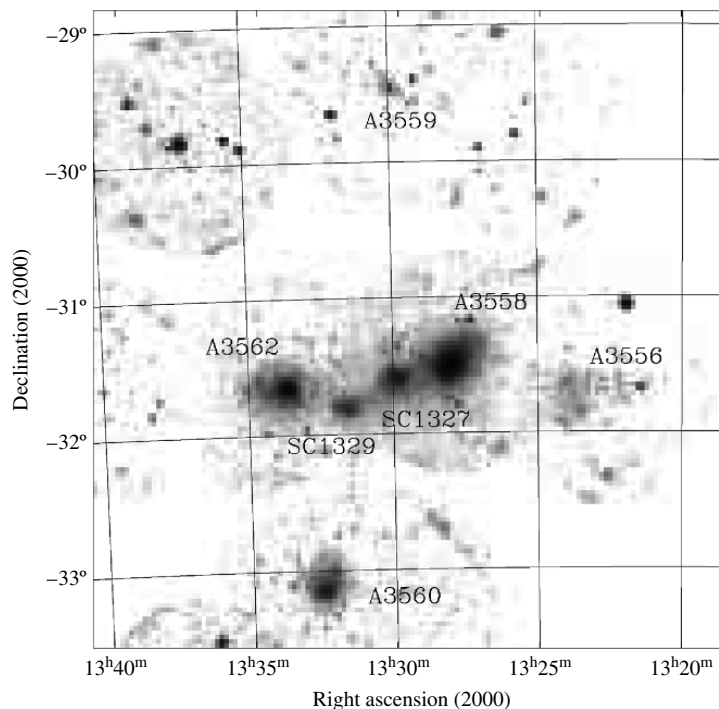


FIGURE 27.20 A large-scale view of the center of the Shapley concentration, centered on the galaxy A3558 and its companions. The entire collection of galaxies is about 10° across in the sky. (Figure from Bardelli et al., *Astron. Astrophys.*, 396, 2002.)

direction of the Local Group's motion and very close to the direction of the Great Attractor. However, its distance of $140h^{-1}$ Mpc is three times the distance calculated for the Great Attractor, so it cannot be identified with the GA. Still, it is probably responsible for 10–15% of the net acceleration of the Local Group.

Astronomers believe that the large-scale bulk motions cannot continue on increasingly larger scales because that would violate the cosmological principle that the universe is isotropic. The direction such a large-scale bulk motion would be a preferred direction in the universe. Recent observations based on Type Ia supernova distances (which have a smaller uncertainty than the Tulley–Fisher distances often used in these studies) indicate that these vast bulk flows eventually converge with the Hubble flow (with no peculiar velocity) at a distance of about $50\text{--}60h^{-1}$ Mpc. In fact, some observations using Type Ia supernovae fail to find any evidence at all of large-scale coherent motion! We will have to wait for more complete and precise observations before the issue can be settled.

Bubbles and Voids: Structure on the Largest Scales

To find how the universe looks when examined at even larger scales, extensive surveys of the sky have been conducted. During the 1950s, the Palomar Observatory Sky Survey recorded the northern portion of the heavens on glass photographic plates, each covering a $6^\circ \times 6^\circ$

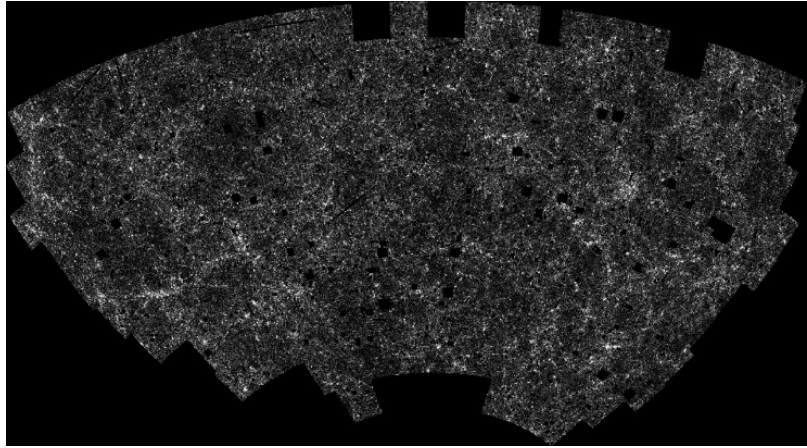


FIGURE 27.21 Two million galaxies, compiled by the United Kingdom Schmidt Telescope sky survey over a region of 4300 square degrees, centered on the south Galactic pole. (Figure from Maddox et al., *MNRAS*, 242, 43p, 1990.)

area. In this study, two plates were made for each area of sky, recorded on emulsions that were more sensitive to either red or blue light, thus giving an indication of the temperatures of the objects photographed in addition to their relative brightnesses. Figure 27.21 shows the result of a more recent survey of the southern skies, made with the United Kingdom Schmidt Telescope. This map displays some two million galaxies between apparent magnitudes 17 and 20.5 that are found within 4300 square degrees of the sky, centered on the south Galactic pole. It is not a photograph; rather, each dot has an intensity (black to white) representing between zero (black) and 20 (white) galaxies. It is obvious that the galaxies are not randomly distributed across the sky. The small, bright regions are clusters, and the longer, filamentary strands are superclusters.

The human eye and brain have a talent for finding patterns, even when none exist. In the case of Fig. 27.21, the difficulty is compounded because it is a two-dimensional depiction of the sky. All of the galaxies are projected onto the plane of the sky, so true associations cannot be distinguished from line-of-sight coincidences. Three-dimensional structures may be hidden or distorted when flattened into two dimensions. In the 1980s, several groups of astronomers began making three-dimensional maps of the universe by using the redshifts of galaxies and the Hubble law to provide the distance of the galaxies. **Redshift surveys** such as these revealed the existence of huge voids, immense versions of the vacant regions seen in Fig. 27.12. These voids measure up to 100 Mpc across and are vaguely spherical, rather than flattened like the sheets occupied by superclusters. The space inside a void is empty of bright spirals and ellipticals, although a few faint dwarf ellipticals may be present.

Most of the early surveys sampled just one area of the sky, so it was not possible to get an overall picture of the distribution of galaxies. To remedy this, Margaret Geller and John Huchra of the Harvard–Smithsonian Center for Astrophysics (CfA), along with their co-workers, carried out surveys of wedge-shaped slices of the heavens; see Fig. 27.22. Like making parallel cuts through a block of Swiss cheese, this technique shows where the empty

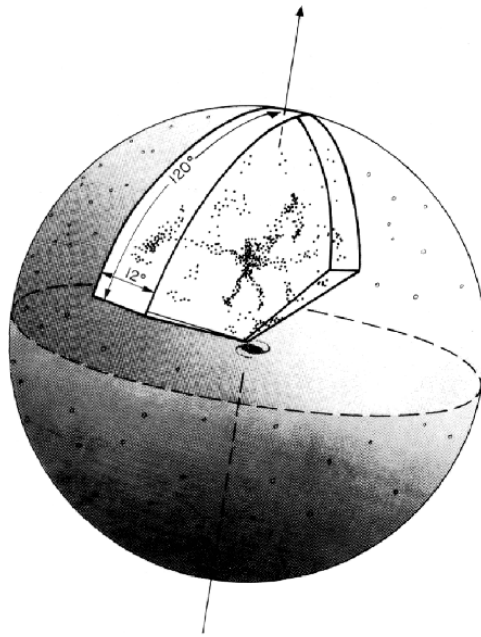


FIGURE 27.22 Two adjacent 6° -wide wedges used in the CfA redshift survey. (Figure from Geller, *Mercury*, May/June 1990.)

regions are and how they relate to one another. During a night's observation, the rotation of Earth carries the telescope through about 9 hours of right ascension, and redshifts are measured for the galaxies that cross the telescope's field of view. This procedure is repeated until all of the galaxies in a 6° declination slice have been measured, out to a radial velocity of some $15,000 \text{ km s}^{-1}$ (a distance of about $150h^{-1} \text{ Mpc}$, according to Hubble's law).³²

Figures 27.23(a) and (b) show two of the first slices, completed in 1985. Because of the uncertainty in the value of H_0 at the time, the distance is expressed in terms of a radial velocity cz , where z is the measured redshift (cf., Eq. 27.8). Each dot on the plots is a galaxy. Study the figures and try to imagine a three-dimensional distribution of galaxies that would have these cross sections. Clearly, the galaxies are not randomly distributed. The slices also show that galaxies do not form a network of threadlike filaments that meander through space. If they did, then a slice would be much more likely to cut across a filament than along it (think of cutting through a bowl of cooked spaghetti). There are simply too many connected features, and too few isolated clumps, for this to be true.

Another terrestrial analogy does supply a structure that matches these observations. A slice through a froth of soap bubbles would look very much like what is seen here. Figures 27.23(a) and (b) show that galaxies lie on two-dimensional sheets that form the walls of bubble-shaped regions of space and that inside the bubbles are huge voids.³³ One

³²For distances over roughly $100h^{-1} \text{ Mpc}$, only the brighter galaxies could be measured, producing a sparser diagram near the curved edge.

³³Another possibility is that the voids are connected like the interior of a sponge, with thin sheets of galaxies separating the passageways.

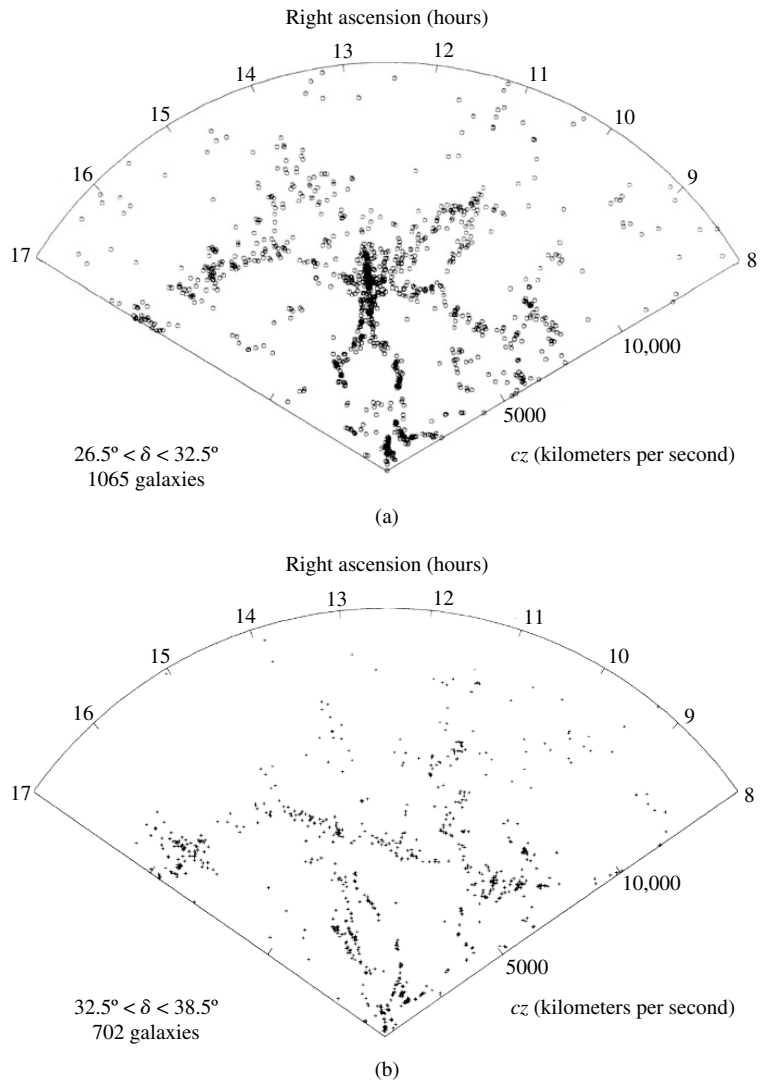


FIGURE 27.23 Two slices from the CfA redshift survey. Right ascension is indicated along the top of each figure. (a) $26.5^\circ < \delta < 32.5^\circ$. (b) $32.5^\circ < \delta < 38.5^\circ$. (Figures adapted from Geller, *Mercury*, May/June 1990.)

such large void is at $\alpha = 15^h$, $v = 7500 \text{ km s}^{-1}$ ($75h^{-1} \text{ Mpc}$ away) and has a diameter of 5000 km s^{-1} ($50h^{-1} \text{ Mpc}$). In Fig. 27.23(a), the Coma cluster is the torso of the humanlike figure at the center of the slice.³⁴ Apparently, rich clusters and superclusters are found at

³⁴The large peculiar velocities of the cluster's galaxies cause them to be spread out in the radial velocity direction. This somewhat elongates the Coma cluster's appearance.

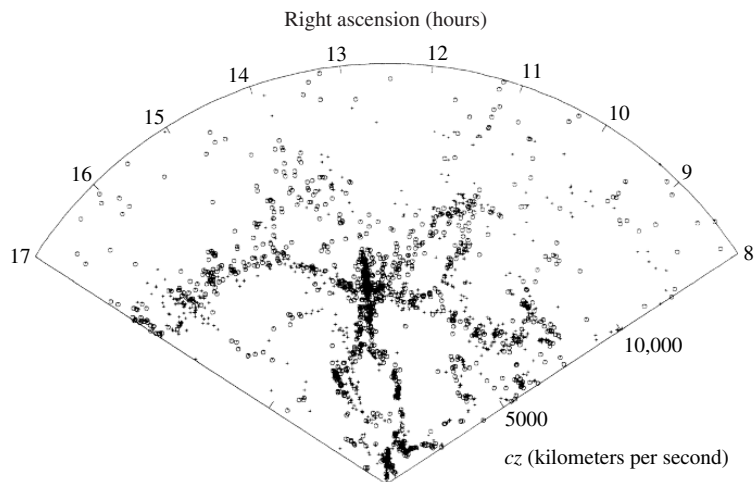


FIGURE 27.24 A combination of the first two slices of the CfA redshift survey, for $26.5^\circ < \delta < 38.5^\circ$. (Figure adapted from Geller, *Mercury*, May/June 1990.)

the intersections of bubbles, and long structures like the Perseus–Pisces supercluster may lie along the surface of adjoining bubbles. The filamentary features show where the survey slices through the sheets of galaxies that form the walls of the voids. This interpretation is strengthened by Fig. 27.24, which shows the combination of the first two slices on the same diagram. Not only do the features show a general correspondence, but they are slightly offset; some features of the combined slice are somewhat thicker than those of the individual slices. This is because the slices are not perpendicular to the walls of the voids. Instead, the curvature of a wall causes adjacent slices to intersect the wall at slightly different locations.

Figure 27.25 shows a combination of results from the CfA redshift survey of the northern skies with a southern survey carried out by L. Nicolaci da Costa and co-workers. The long arc at the top is the Great Wall, with its counterpart, the Southern Wall, running diagonally across the bottom. At the time of this writing, these are the largest known structures in the universe, and they may consist of the walls of several adjacent bubbles. The Great Wall passes through the arms of the homunculus in Fig. 27.23. It extends for *at least* $150h^{-1}$ Mpc in right ascension and *at least* $60h^{-1}$ Mpc in declination (perpendicular to the diagram) but is less than $5h^{-1}$ Mpc thick. The Great Wall disappears into regions of the sky that are obscured by dust (on one end) or not yet mapped (on the other end). It may actually continue on to join up with the Southern Wall, which contains the Perseus–Pisces supercluster.

The deepest survey to date is the 2dF Galaxy Redshift Survey (2dFGRS) carried out by the Anglo-Australian Observatory. It produced a three-dimensional map of 220,929 galaxies brighter than $m = -19.5$ and covering 5% of the sky. The survey penetrated out to approximately 1 Gpc, to a redshift of $z \approx 0.3$. Figure 27.26 shows a 3° slice that cuts through the north (left) and south (right) galactic poles. It includes 63,000 galaxies and clearly shows the galaxies collecting on the walls of immense voids. The spongelike structure of the universe is apparent in Fig. 27.27, which illustrates how the clusters and superclusters of galaxies clump together.

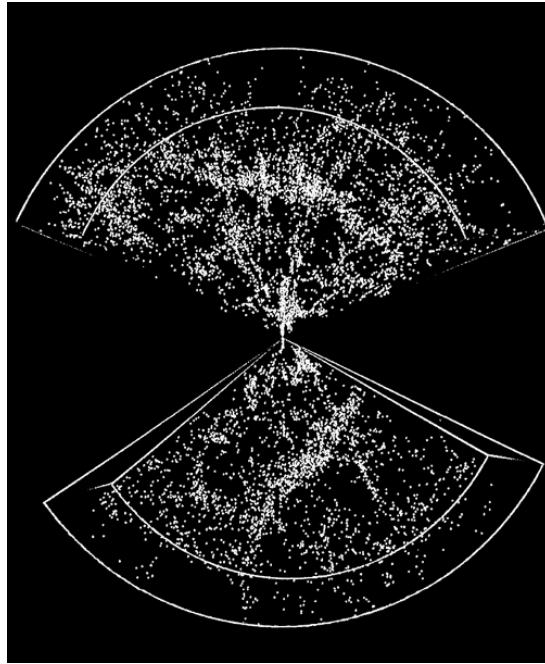


FIGURE 27.25 A cross section of the universe with $cz \leq 12,000 \text{ km s}^{-1}$ (within $120h^{-1}$ Mpc of Earth), showing 9325 galaxies. The coordinates of the top slice are $8^{\text{h}} < \alpha < 17^{\text{h}}$ and $8.5^{\circ} < \delta < 44.5^{\circ}$, those of the bottom slice $20.8^{\text{h}} < \alpha < 4^{\text{h}}$ and $-40^{\circ} < \delta < 2.5^{\circ}$. (Figure from da Costa et al., *Ap. J. Lett.*, 424, L2, 1994.)

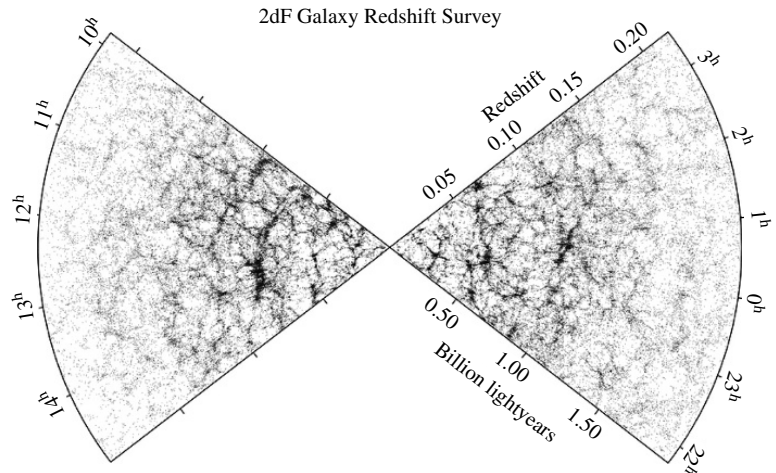


FIGURE 27.26 A 3° slice through the 2dF Galaxy Redshift Survey. It displays 62,559 galaxies. The north galactic pole is on the left, and the south galactic pole is on the right. (Figure adapted from Colless et al., arXiv:astro-ph/0306581, 2003.)

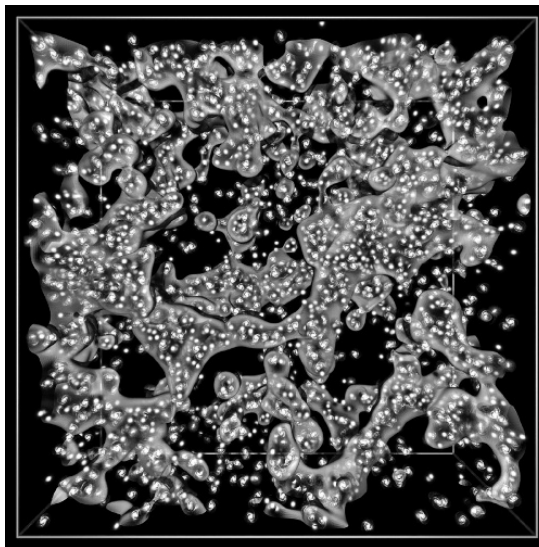


FIGURE 27.27 A cubical volume of space with sides 100 Mpc long. The clusters and superclusters of galaxies within are joined together, illustrating the spongelike structure of space. (Figure courtesy of the 2dF Galaxy Redshift Survey Team.)

The ongoing Sloan Digital Sky Survey (SDSS) is the most recent advance in deep surveys of the universe. The first phase of the project, completed in June 2005, imaged more than 20% of the sky at five wavelengths and recorded some 200 million objects. The SDSS obtained spectra of over 675,000 galaxies, 90,000 quasars, and 185,000 stars. The SDSS utilizes a 2.5-meter telescope on Apache Point, New Mexico. Its 120-megapixel camera can photograph 1.5 square degrees of sky at a time (eight times the area of the full moon). When completed, the SDSS will have produced a three-dimensional map of more than 25% of the sky. Such an extensive map should provide a more coherent and detailed view of the large-scale structures of the universe and supply clues concerning their origin. Figure 27.28, based on the 2dFGRS and SDSS catalogs, shows the clusters of galaxies belonging to the Pisces–Cetus supercluster of galaxies. The lines connecting the 19 clusters of galaxies (at an average redshift of $z = 0.0591$) outline the cosmic web of interconnected filaments that make up this supercluster.

Quantifying Large-Scale Structure

Another method of inferring the presence of large features is to use the available data to mathematically describe the clustering of galaxies. If galaxies were uniformly distributed through space with a number density n , then the probability dP of finding a galaxy within a volume dV would be the same everywhere, $dP = n dV$.³⁵ However, galaxies are not scattered about randomly. The probability of finding a galaxy within a volume dV at distance

³⁵We assume that dV is sufficiently small enough that $dP \leq 1$.

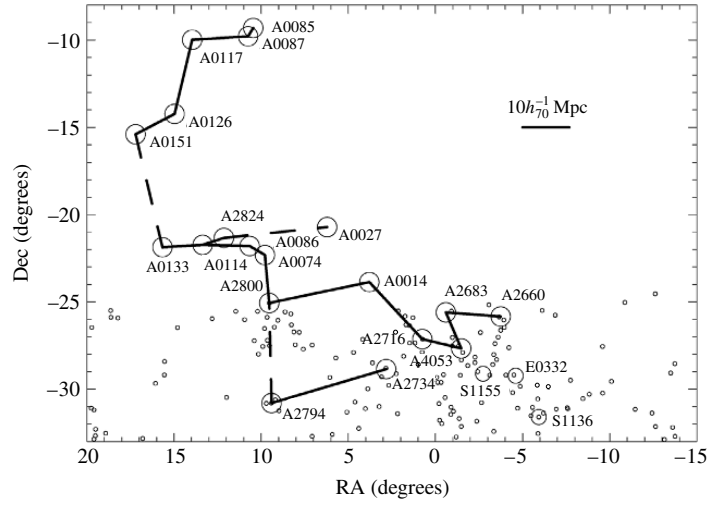


FIGURE 27.28 Clusters of galaxies belonging to the Pisces-Cetus supercluster of galaxies. The lines connecting the clusters are between $20h^{-1}$ Mpc and $25h^{-1}$ Mpc long. (Figure adapted from Porter and Raychaudhury, *Mon. Not. R. Astron. Soc.*, 364, 1387, 2005.)

r from a specified galaxy then becomes

$$dP = n[1 + \xi(r)] dV, \quad (27.21)$$

where n is the average number density of galaxies and $\xi(r)$ is a *two-point correlation function* that describes whether the galaxies are more concentrated ($\xi > 0$) or more dispersed ($\xi < 0$) than average. For 118,149 galaxies from the SDSS and separations of $0.1h^{-1}$ Mpc $< r < 16h^{-1}$ Mpc, the **correlation function** is observed to be of the form

$$\xi(r) = \left(\frac{r}{r_0}\right)^{-\gamma}, \quad (27.22)$$

where r_0 , the correlation length, is $r_0 = 5.77h^{-1}$ Mpc, and $\gamma = 1.8$. The correlation length varies with the luminosities of the galaxies and ranges from $4.7h^{-1}$ Mpc to $7.4h^{-1}$ Mpc, brighter galaxies having the longer correlation lengths.

As surveys such as the SDSS sample more remote regions of space (say, within a sphere of radius R_s), the behavior of r_0 with sample depth R_s is of some interest. If, as some have suggested, the universe has a fractal structure and is scale-free, then r_0 should increase proportionally with R_s . (If the universe looks essentially the same at all scales, then deeper observations must reveal ever larger versions of smaller structures.) However, the cosmological principle of homogeneity requires that r_0 become constant for sufficiently deep samples, because beyond some R_s there are no larger structures to be found.

The search is under way to find the value of R_s where the universe becomes homogeneous. Although some studies, such as the one shown in Fig. 27.29, have apparently found the transition to homogeneity, others have not. There is general agreement that the universe

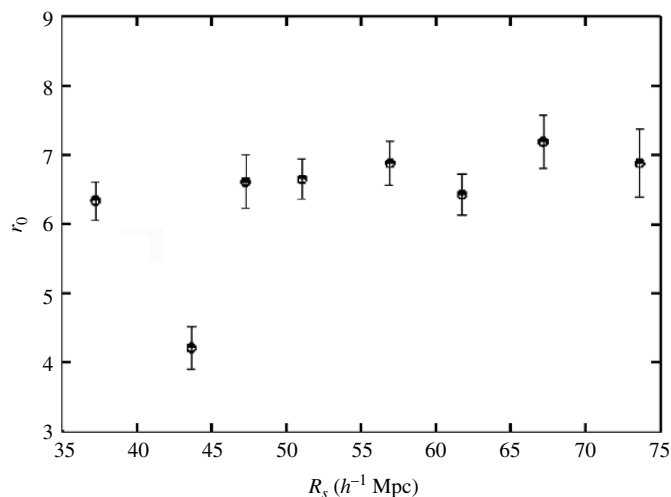


FIGURE 27.29 The correlation length r_0 as a function of the sample depth R_s for the CfA-II catalog (a galactic redshift catalog compiled by the Harvard Center for Astrophysics). The flat plateau on the right indicates a transition to homogeneity for this study at about $R_s \approx 60\text{--}70h^{-1}$ Mpc. (Figure adapted from Martinez et al., *Ap. J.*, 554, L5, 2001.)

has a fractal structure on scales out to $\approx 30h^{-1}$ Mpc. (This does *not* mean that the universe is a fractal, because the self-similarity holds only for smaller scales and not for larger ones.) The prevailing view is that the transition to homogeneity occurs somewhere around or past $R_s \approx 30h^{-1}$ Mpc, perhaps as far as $R_s \approx 200h^{-1}$ Mpc. This scale is consistent with the extent of the largest coherent structures in the universe, the cosmic network of interconnected filaments of galaxies that fill the universe (see Fig. 27.28).

What Were the Seeds of Structure?

It is not easy to imagine how the pattern of galaxies revealed by the redshift surveys could have come about from an initially random distribution of galaxies. By chance, of course, some areas would have had more galaxies than others. The denser regions would have gravitationally attracted still more galaxies, leaving the less dense volumes empty. Although gravity does act to increase the difference between high- and low-density regions, it is not enough to explain the voids. The large voids are some 100 Mpc across, but the peculiar velocities of galaxies are only on the order of hundreds of kilometers per second. Using a characteristic peculiar velocity of $\sim 600 \text{ km s}^{-1}$, the time for a galaxy to cross a void is at least 160 Gyr. This is much longer than the age of the universe, t_H , and means that the galaxies could *not* have formed inside the voids and then moved out. Instead, the galaxies must have formed nearly in their present pattern. Apparently, the seeds of the structure we see today were present from the earliest moments of the universe. The question of the origin of this amazing arrangement of galaxies will be taken up again in Chapter 30.